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Paleomagnetism of the Resurrection Peninsula, Alaska: Implications for the Tectonics of Southern Alaska and the Kula-Farallon Ridge

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The rocks of the Resurrection Peninsula compose an ophiolite (57±1 Ma) within the vast accretionary prism of the Chugach-Prince William terrane. Paleomagnetic data from pillow basalt and sheeted dikes of the ophiolite yield a mean paleolatitude of 54°±7°, which implies 13±9° of poleward displacement. The characteristic component of the remanence resides in a Ti-poor titanomagnetite with a blocking temperature close to 580°C. A positive fold test and the petrology of the magnetic minerals lead us to conclude that the characteristic remanent magnetization (ChRM) in the sheeted dikes originated as a primary thermoremanent magnetization (TRM). We hypothesize that the ChRM of the pillow basalt originated as a chemical remanent magnetization (CRM) acquired soon after formation of the rocks during hydrothermal metamorphism. Structural data indicate polyphase deformation with folding about vertical and horizontal axes. The paleomagnetic data pass a fold test only when we incorporate this interpretation in our structural correction. Geologic relationships lead us to interpret the Resurrection Peninsula ophiolite as a fragment of the extinct Kula-Farallon ridge that accreted to the Chugach-Prince William terrane soon after formation. The paleomagnetic paleolatitude indicates that the ridge intersected the North American continental margin in the vicinity of northern Washington in late Paleocene time. The paleostrike of the sheeted dikes (N26°E±12°) shows that the ridge trended NNE. According to current plate tectonic models, if the Chugach-Prince William terrane moved north with the full coastwise displacement of the Kula plate, the Resurrection Peninsula ophiolite would have arrived at its present position at 45 Ma. This age is perhaps slightly later than that of the terranes that lay immediately inboard (Peninsular and Wrangellia terranes), indicating some relative displacement between those terranes and the Chugach-Prince William terrane. Such displacement, in turn, agrees with studies which imply a Coast plutonic-metamorphic complex provenance for the Chugach-Prince William terrane.

INTRODUCTION

Most geoscientists now accept that southern Alaska comprises a collage of tectonostratigraphic terranes [*Coney et al.*, 1980; *Jones et al.*, 1983]. While many of these terranes are allochthonous, the majority of geologic and paleomagnetic evidence supports the hypothesis that most of them were nearly in place (with respect to the rest of North America) by Late Cretaceous or early Tertiary time [*Hillhouse and Grommé*, 1982; *Hillhouse et al.*, 1985; *Thrupp and Coe*, 1986; *Coe et al.*, 1985; *Globerman*, 1985; *Panuska et al.*, 1990; *Csejtey et al.*, 1982; *Nokleberg et al.*, 1985; *Wallace et al.*, 1989]. For the outboard Chugach and Prince William terranes, however, paleomagnetic data indicate large Cenozoic poleward translation [*Hillhouse and Grommé*, 1977; *Stone et al.*, 1982; *Plumley et al.*, 1983], contradicting geologic data which tie these terranes to their more stationary, inboard neighbors [*Moore et al.*, 1983; *Davies and Moore*, 1984; *Plafker*, 1987; *Plafker et al.*, 1989]. *Coe et al.* [1985] screened southern Alaska paleomagnetic studies according to reliability (e.g., fold test, reversal test, lithology) but still found that the most reliable of these studies, from the Paleocene Ghost Rocks Formation [*Moore et al.*, 1983], shows significant poleward translation [*Plumley et al.*, 1983].

We studied the paleomagnetism of early Tertiary pillow basalts and sheeted diabase dikes from the Resurrection Peninsula ophiolite, Alaska, in order to determine its paleolatitude for comparison with other results from the Chugach-Prince William terrane. A preliminary study [*Hillhouse and Grommé*, 1977; *Grommé and Hillhouse*, 1981] indicated 24°±12° of poleward displacement, in close agreement with the 25°±7° of poleward displacement determined by *Plumley et al.* [1983] for pillow volcanic rocks from Kodiak Island. However, due to poor age constraints and ambiguities in the structural correction, many workers regarded the Resurrection Peninsula result with suspicion. In the present study, we resolve the structural ambiguity and take advantage of a new, tightly constrained, U/Pb age [*Nelson et al.*, 1989]. We combine new data from extensive sampling and detailed measurement with data from *Hillhouse and Grommé* [1977] and obtain a paleomagnetic pole that agrees closely with the previous preliminary result. The age of the Resurrection Peninsula ophiolite, however, is younger than previously thought (circa 57 Ma, rather than 70 Ma) and requires considerably less poleward displacement than *Hillhouse and Grommé* [1977] reported (13°±9°, rather than 24°±12°).

The Resurrection Peninsula ophiolite is set within the large, laterally extensive flysch belt of the Chugach-Prince William

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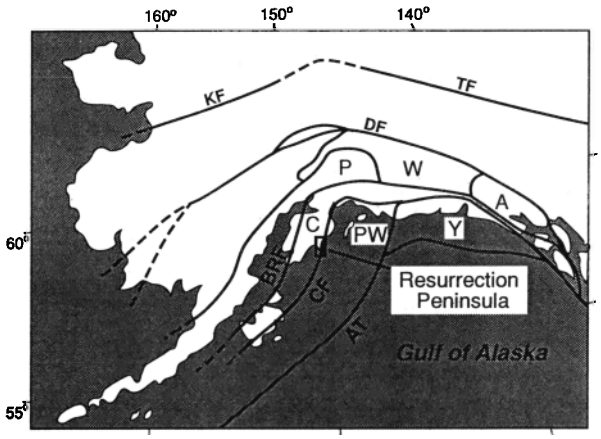


Fig. 1. Map of southern Alaska showing major terranes and faults [from *Plafker*, 1987] with location of Figure 2, the Resurrection Peninsula. Terranes are A, Alexander; C, Chugach; P, Peninsular; PW, Prince William; W, Wrangellia; and Y, Yakutat. Faults are AT, Aleutian trench; BRF, Border Ranges fault; CF, Contact fault; DF, Denali fault; KF, Kaltag fault; and TF, Tintina fault.

terrane (Figure 1), a long-lived accretionary complex that ranges in age from Early Cretaceous through Paleogene. Previous workers subdivided the accretionary belt into two terranes [*Plafker et al.*, 1977] based on relations in the northeastern Gulf of Alaska region. The Chugach terrane comprises the Cretaceous, inboard part of the complex, while the Prince William terrane includes only the Tertiary, outboard part. The Contact fault system juxtaposes these two terranes and, though usually easily identified in the eastern Gulf of Alaska region [*Winkler and Plafker*, 1975, 1981; *Winkler et al.*, 1981], in the western Gulf of Alaska region it is difficult to discern which of many faults compose the Contact fault system [*Tysdal and Case*, 1979; *Nelson et al.*, 1985; *Bol and Coe*, 1987; *Dumoulin*, 1987, 1988; *Gibbons*, 1988; Bol and Gibbons, in press]. In the eastern gulf, Chugach terrane rocks are generally more deformed and metamorphosed than the Prince William terrane rocks. In the western gulf, complex lithologic and structural relationships, a paucity of age control, and no distinctive petrographic breaks combine to obscure the terrane boundary. Due to this ambiguity in terrane distinction in the western gulf, we herein treat the two terranes as one: the Chugach-Prince William terrane.

Geologic Setting

The Resurrection Peninsula ophiolite is a west dipping sequence of (in descending order) pillow basalt with minor interbeds of thin clastic turbidites, sheeted dikes, isotropic gabbro, and layered gabbro floored by a highly sheared sequence of sedimentary rocks with intercalated pods of serpentinized ultramafic rocks (Figure 2) [*Tysdal et al.*, 1977; *Nelson et al.*, 1985]. *Tysdal et al.* [1977] originally mapped the contact between the ophiolite and the adjacent flysch as depositional, in part. They also reported dikes intruding surrounding flysch and local pillow basalt interbedded with basaltic tuff on Renard Island, immediately west of the Resurrection Peninsula. The tuff is, in turn, interbedded with the surrounding flysch. Recent, more detailed mapping by *Nelson et al.* [1985] revealed that the Resurrection Peninsula ophiolite appears to lie everywhere in fault contact with the adjacent flysch and tuff. Furthermore, the flysch on Renard Island was metamorphosed to biotite-grade greenschist facies while the flysch interbedded with pillow basalt on the peninsula shows no sign of metamorphism. *Nelson et al.* [1985] did not observe pillows interbedded with tuff on Renard Island.

In addition to sampling, we mapped the contact and structural relations between the pillow basalt and sheeted dikes. Diabase dikes range from 0.1 to 5 m thick, usually 0.5-2 m, only locally crosscutting one another. Although most dikes we observed have two chilled margins, many have only one. Estimates of paleohorizontal (i.e., bedding and pillow tops) for the dikes were obtained either from small sections of pillows trapped within the dikes or from nearby attitudes of pillow basalt. A gradational zone about 10 m thick separates the sheeted dikes and pillow basalt.

Pillow basalts of the Resurrection Peninsula crop out along the western shore in excellent, fresh, wave-cut exposures. Pillows are 0.2-2 m thick, 0.5-5 m long, and occur in pillowed flows that are usually 3-10 m thick and commonly have a massive base. Pillow aspect ratios range from nearly spherical to highly flattened (5:1 width:thickness), generally increasing with height within an individual flow and with their long dimension usually downdip [*Tysdal et al.*, 1977]. We obtained estimates of the paleohorizontal at two localities from interbedded, thin, silty turbidites, whereas the three remaining

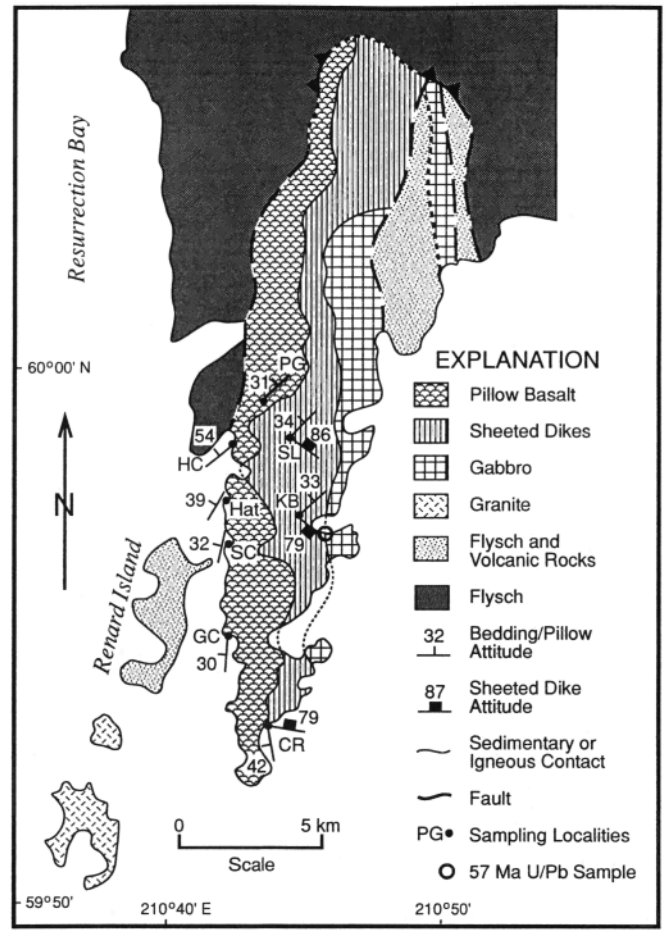


Fig. 2. Geologic map of the Resurrection Peninsula [after *Tysdal et al.*, 1977; *Nelson et al.*, 1985] with sampling localities and structural data from this study. Localities are CR, Cape Resurrection; GC, Good Cove; Hat, Hat Cove; HC, Humpy Cove; KB, Killer Bay; PG, Porcupine Glacier; SC, Sandspit Cove; and SL, Spud Lake.

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pillow localities rely on averaged attitudes from pillows with high aspect ratios.

Metamorphism

The ophiolitic rocks have been extensively metamorphosed to the lower greenschist facies. Analysis of 12 thin sections distributed over all sampling localities shows common sericite replacement of plagioclase, chlorite replacement of pyroxene, and minor amounts of metamorphic epidote, quartz, albite, pumpellyite, prehnite, calcite, and serpentinite. *Alt et al.* [1985] described a similar metamorphic assemblage in Deep Sea Drilling Project (DSDP) hole 504B, the only core to penetrate oceanic crust to the sheeted dike complex. They ascribed the alteration to hydrothermal metamorphism at or near a spreading ridge. In their study, fluid inclusion and oxygen isotope data indicated metamorphic temperatures of 250°-380°C. *Schöps and Herzig* [1990] reported somewhat higher temperatures (350°-500°C) for dikes deeper within hole 504B.

Age

Tysdal et al. [1977] inferred a Cretaceous age for the Resurrection Peninsula ophiolite on the basis of an interpreted igneous intrusive contact with the adjacent flysch. They assigned a Cretaceous age to the flysch on the basis of fossils found along strike tens of kilometers away combined with middle greenschist facies metamorphism and polyphase deformation. As mentioned above, new mapping shows that the ophiolite lies in fault contact with the surrounding flysch. Thus the inferred age of the flysch has no bearing on the age of the igneous rocks. *Nelson et al.* [1989] recently obtained a U/Pb (zircon) age of 57±1 Ma for a plagiogranite on the Resurrection Peninsula. The plagiogranite intrudes the dikes and is, in turn, cut by dikes and thus dates the sheeted dike complex (and by association the pillow basalts). M. L. Silberman (unpublished data reported by *Winkler et al.* [1984]) obtained three whole rock, K/Ar ages of 46±2, 47±2, and 50±2 Ma on dike samples but considered them suspect because (1) the ophiolite was thought to lie in depositional contact with the adjacent Valdez Group and (2) a similar age is reported for regional greenschist facies metamorphism [e.g., *Plafker*, 1987]. Given the uncertainties in K/Ar dating, much less the whole rock dates on oceanic rocks, we feel the K/Ar ages provide only a minimum age of magnetization.

The age spanned by the ophiolite is probably of the order of a few hundred thousand years. As an approximation, if we assume that the ophiolite is an accreted fragment of the Kula-Farallon ridge (discussed below) and that it originated near the continental margin at 54°N (its mean paleolatitude), using the Euler poles of *Engebretson et al.* [1985], the spreading rate would be 60 km per million years (60 mm/yr). North to south, the ophiolite is 30 km long. If all the dikes were oriented east-west, the dikes would require 500,000 years of ridge spreading for emplacement.

The age of deformation of the ophiolite is uncertain. There are no crosscutting or overlap relationships to provide constraints. In the surrounding Valdez Group flysch, circa 57 Ma dikes that cut the accretionary prism are deformed only by some late stage faults [*Bradley and Kusky*, 1991]. Therefore most deformation of the accretionary prism was concluded by that time. It seems reasonable to surmise then that the ophiolite was deformed during accretion and little hence. However, we have no direct evidence to prove this contention.

The 57-Ma age (late Paleocene [*Harland et al.*, 1990]) calls into question the identification of the Contact fault because the fault, which separates Cretaceous from Tertiary rocks, was previously mapped to the east, or seaward, of the Resurrection Peninsula. If the Resurrection Peninsula ophiolite is exposed as a fenster within Cretaceous rocks, then the fault between the ophiolite and the flysch must be the Contact fault. The ambiguity surrounding the identification of the Contact fault supports our decision to group the Chugach and Prince William terranes together for the purposes of this paper.

PALEOMAGNETIC SAMPLING AND LABORATORY TECHNIQUES

Hillhouse and Grommé [1977] sampled the pillow basalt at Good Cove and sheeted dikes at localities near Killer Bay and Cape Resurrection obtained one core per dike. The pillow basalt at Good Cove was sampled in more detail by A. J. Bol who obtained 4-8 cores per site and expanded the sampling to another two localities in dikes (Spud Lake and Humpy Cove) and four localities in pillow basalt (Porcupine Glacier, Humpy Cove, Hat Cove, and Sandspit Cove). We oriented cores with a Brunton compass and sighted landmarks from the orientation stage to correct for spurious magnetic declination anomalies. These anomalies were less than 5° and we applied the correction to both paleomagnetic and structural data.

All samples were measured on cryogenic magnetometers at the U.S. Geological Survey, Menlo Park, California, and University of California at Santa Cruz (where the magnetometer is located in a very low field laboratory). *Hillhouse and Grommé* [1977] employed stepwise alternating field (af) demagnetization on pilot samples and an abbreviated procedure with 2-3 demagnetization steps for the rest of the samples. Bol treated pilot samples with both thermal and af stepwise demagnetization and thermally demagnetized the remainder of the cores in a series of 10-16 steps.

PALEOMAGNETIC RESULTS

Sheeted Dikes

The sheeted dikes exhibited fairly uniform magnetic characteristics over the study area. The intensity of natural remanent magnetization (NRM) ranges from 4×10^{-7} to 7×10^{-3}

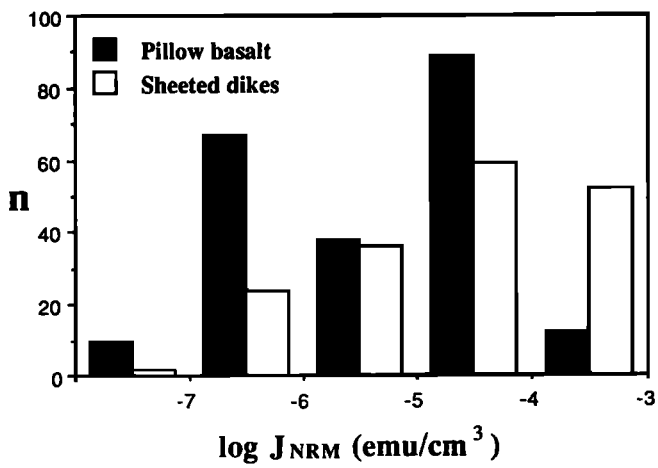


Fig. 3. Histogram of natural remanent magnetization (NRM) intensities for pillow basalt and diabase dike samples. While the range of intensities for the two is similar, dike samples tend to have stronger intensities.

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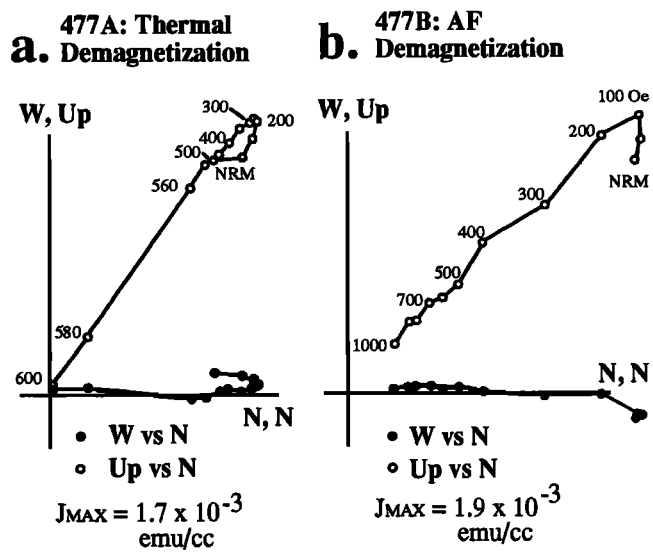


Fig. 4. Orthogonal vector diagrams (in situ coordinates) of companion specimens from a diabase dike sample given (*a*) thermal and (*b*) alternating field (af) stepwise demagnetizations. The two specimens show nearly identical directions. Also note the large relative magnitude of the characteristic magnetization to the secondary overprint, typical of the diabase dike samples.

emu/cm^3 and average 10^{-3} to 10^{-4} emu/cm^3 (Figure 3). Stepwise demagnetization removed a weak secondary component (absent from 74% of specimens) before isolating the characteristic component (Figure 4). Vastly different directions distinguish the secondary component (north and steeply down, in situ) from the characteristic component (northwest and steeply up). Usually, thermal demagnetization isolated the characteristic direction by 400°C, but in 30% of the samples, the secondary component persisted to temperatures greater than 500°C (Figure 5). Intensity remained high throughout until a precipitous drop at temperatures over 550°C. Alternating field demagnetization usually isolated the primary component at peak fields of 200 Oe or less (Figure 6). The median destructive field (MDF) ranged broadly from 100 to 900 Oe (Figure 7). Thermal and alternating field treatments gave nearly the same characteristic directions for companion specimens of the same core (Figure 4).

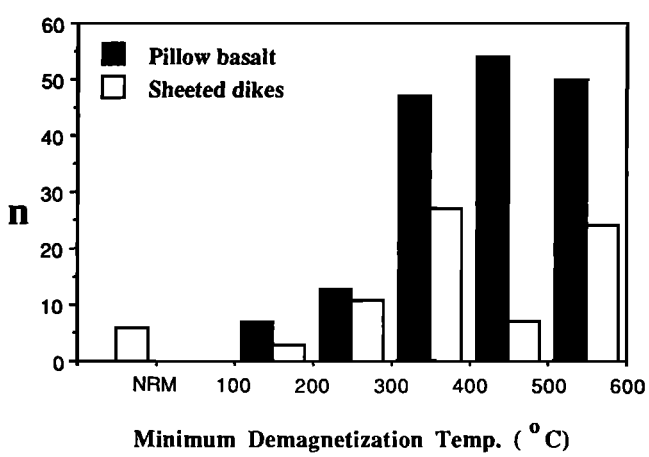


Fig. 5. Histogram of minimum unblocking temperature necessary to reveal characteristic remanent magnetization (ChRM) for pillow basalt and diabase dike samples. Unblocking temperatures frequently exceed 500°C.

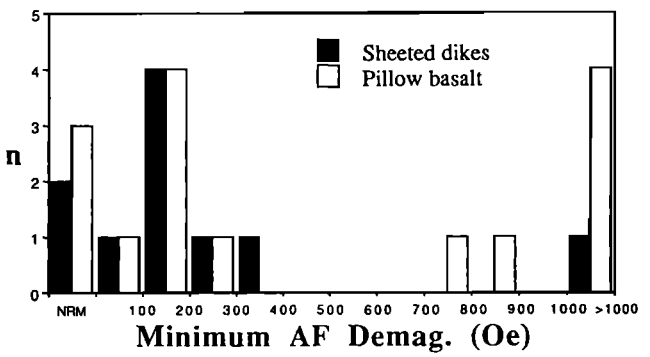


Fig. 6. Histogram of minimum af demagnetization required to reveal the characteristic component in pillow basalt and diabase dike samples. The coercivity of the overprint occasionally exceeded 1000 Oe, especially in pillow basalt.

Pillow Basalt

Magnetic characteristics of pillow basalts differed markedly from those of the dikes. The main difference arose from the relative magnitude of the secondary component, which dominated the NRM of pillow basalt samples (Figure 8). NRMs showed the same range (4 x 10^{-7} to 3 x 10^{-3} emu/cm^3) but were somewhat weaker and displayed more variability between localities (Figure 3). Median destructive field is more variable, ranging from 50 to greater than 1000 Oe (Figure 7). Component directions were similar to those for dikes, and the secondary component in pillows usually persisted to higher temperatures (Figure 5). Alternating field demagnetization commonly removed the secondary component by 200 Oe (Figure 6) but occasionally failed completely to reveal the characteristic component (Figure 8). Therefore in the analysis of the component directions for pillow basalt, only thermal demagnetization data were used. This decision resulted in little loss of data, though; it affected only data from pilot specimens with thermally demagnetized companions and one site from *Hillhouse and Grommé* [1977], which was reoccupied in the more recent study.

Rock Magnetism

Blocking temperatures between 500° and 580°C observed in rocks of the Resurrection Peninsula suggest that the remanence may reside in Ti-poor titanomagnetite. Isothermal remanent magnetization (IRM) studies show magnetic saturation at fields between 2 and 5 kG, supporting this conclusion (Figure 9).

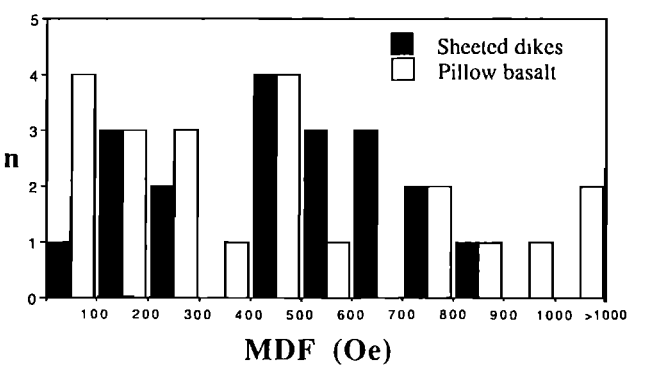


Fig. 7. Histogram of median destructive field (MDF) for pillow basalt and diabase dike pilot samples given stepwise AF demagnetization. Some pillow basalt MDFs exceeded 1000 Oe.

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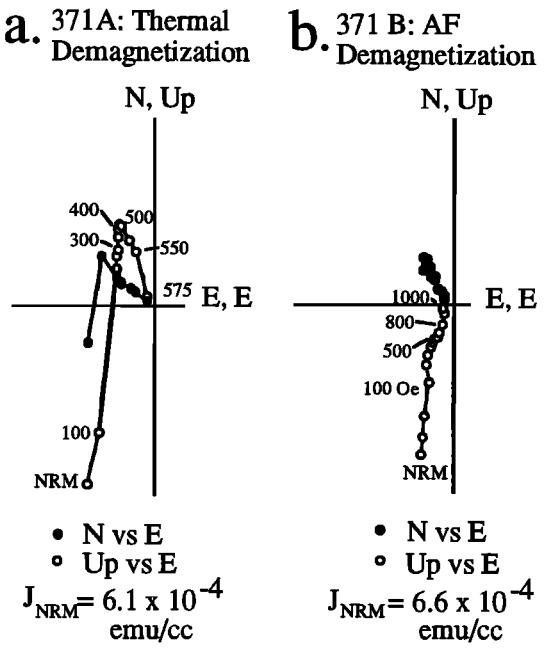


Fig. 8. Orthogonal vector diagrams (in situ coordinates) of companion specimens from a pillow basalt sample given (*a*) thermal and (*b*) af stepwise demagnetizations. The secondary overprint dominates the NRM of both specimens. In this sample, thermal demagnetization successfully revealed the characteristic component, while af demagnetization did not.

Conversely, the magnetization of pillow basalts from dredge hauls and DSDP cores resides in titanomaghemite [e.g., *Petersen et al.*, 1979; *Johnson and Hall*, 1978; *Prévot et al.*, 1979]. A product of low-temperature oxidation on or beneath the seafloor, titanomaghemite has a lower spontaneous magnetization and Curie temperature than does magnetite. At temperatures of 250°-300°C, titanomaghemite inverts to Ti-poor titanomagnetite and ilmenite. So how can the remanence of the Resurrection Peninsula ophiolite be primary when it clearly resides in a secondary mineral phase? We believe the answer lies in the magnetization of the sheeted dike complex.

In thin sections from the sheeted dikes, we commonly observed a relict trellis pattern, indicating oxidation/exsolution of ilmenite and magnetite from titanomagnetite. Hematite partially replaces the ilmenite lamellae, corresponding to stage C4-C5 oxidation of *Haggerty* [1976]. Ilmenite lamellae usually form during high-temperature oxidation of titanomagnetite during primary cooling of the rock. In that case, the titanomagnetite disproportionates, usually before it cools through its blocking temperature spectrum and acquires a thermoremanent magnetization (TRM) that resides in Ti-poor titanomagnetite.

Alternatively, a rock may be oxidized to titanomaghemite at low temperature, then be heated to 250°-300°C and invert to Ti-poor titanomagnetite with ilmenite lamellae [*Inoue*, 1982]. *Inoue* [1982] reported that the lamellae formed by this process commonly appeared as ilmenohematite/hemoilmenite sandwiches. We did not observe this sandwich structure in our study.

As mentioned previously, the silicate minerals in the rocks show alteration consistent with hydrothermal metamorphism at a spreading ridge. Titanomagnetites subjected to such metamorphism develop a granular or cloth-like texture [*Ade-Hall et al.*, 1971; *Facey et al.*, 1985], unless they had already oxidized into Ti-poor magnetite with ilmenite lamellae. The magnetite grains from sheeted dikes show no granular or cloth-like texture but rather display lamellae consistent with high-temperature oxidation. From this observation, we conclude that the titanomagnetite was oxidized before the dikes underwent hydrothermal metamorphism. Therefore the petrographic evidence supports the hypothesis that the remanence residing in the Ti-poor magnetite of the sheeted dikes was acquired as a TRM, as opposed to a late stage chemical remanent magnetization (CRM).

Thin sections from pillow basalts revealed no titanomagnetite; we observed only pyrite as an opaque phase. The remanence must reside in submicroscopic grains. Because the metamorphism was nearly the same in both the dikes and pillows, and paleomagnetic directions do not differ significantly between the two, we hypothesize an early acquisition of remanence for pillows as well as dikes. However, this remanence most likely was acquired as a CRM early in the history of the rocks. We envision a history where extrusion and quenching of pillow basalt with typical titanomagnetite (x = 0.6) was followed by low-temperature oxidation to produce titanomaghemite. Subsequent phase splitting to ilmenite + Ti-poor magnetite during hydrothermal metamorphism led to a CRM residing in the magnetite. Parallel directions from dikes and pillows indicate that maghemitization, metamorphism, and phase splitting (and thus acquisition of remanence) occurred soon after formation of the rocks. *Facey et al.* [1985] hypothesized a similar history for the origin of Ti-poor titanomagnetite observed in pillow basalt in DSDP hole 504B.

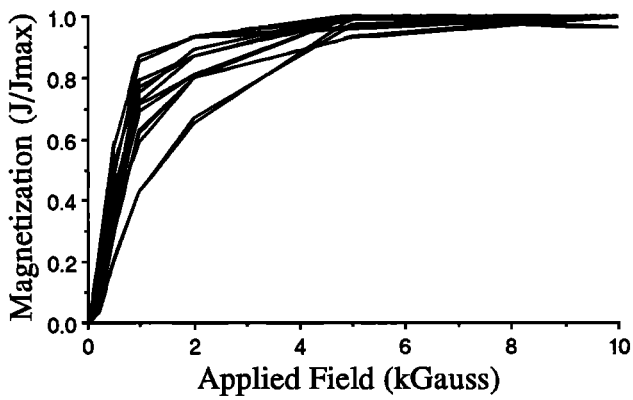


Fig. 9. IRM plot of 12 representative samples taken from throughout the study area. Rapid increase in IRM acquisition in fields up to 1000 G and saturation between 2 and 5 kG indicate Ti-poor titanomagnetite as the dominant magnetic mineral.

Analysis of Data

Secondary Component

We used only samples that had been stepwise demagnetized in our analysis of the secondary component. We plotted the data on orthogonal vector diagrams [*Zijderveld*, 1967] and obtained directions using a least squares fit [*Kirschvink*, 1980] to three or more demagnetization steps. Many dike samples, however, exhibited very little secondary overprint, and therefore pillow basalts provide most of the data. We combined sample directions into site mean directions, and all site mean directions into a mean for the Resurrection Peninsula ophiolite

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TABLE 1. Secondary Component, Site Mean Directions From Dikes and Pillow Basalt, Resurrection Peninsula, Alaska.

				In Situ		Tilt Corrected	
Site	N/n	k	α_{95}	Dec	Inc	Dec	Inc
R1	7/7	22.3	13.1	356.0	69.6	337.7	17.8
R3	5/6	17.3	18.9	343.7	73.5	326.0	40.9
R4	3/6	24.7	25.4	14.3	65.5	344.1	38.9
R5	6/8	4.9	33.6	149.6	82.1	14.7	51.1
R7	6/6	11.2	20.9	16.4	66.1	11.1	21.3
R8	8/8	228.5	3.7	353.6	74.1	331.6	22.0
R9	6/6	11.8	20.4	337.0	72.7	327.4	19.1
R10	6/6	73.4	7.9	3.8	78.2	331.6	26.7
R11	3/6	118.0	11.4	68.6	74.9	348.6	60.3
R12	6/6	409.7	3.3	359.6	69.7	336.6	41.4
R15	6/6	39.0	10.9	6.9	78.1	332.7	49.6
R21	6/6	67.8	8.2	9.6	81.8	329.5	52.8
R22	4/6	101.5	9.2	7.6	78.2	332.8	49.8
R23	7/7	69.0	7.3	347.8	81.9	325.2	51.3
R24	5/6	103.0	7.6	23.3	78.2	336.4	52.1
R26	4/6	64.3	11.5	44.1	76.7	342.6	55.0
R28	5/6	210.4	5.3	352.9	75.7	330.5	45.9
R29	5/5	247.0	4.9	1.5	79.3	330.2	50.0
R33	4/6	58.2	12.1	10.7	71.9	308.9	57.2
R34	2/6	N/A	N/A	338.6	71.5	299.8	48.9
R37	5/7	21.8	16.7	33.0	80.3	293.3	63.4
R38	4/6	36.7	15.4	335.0	67.9	313.5	32.1
R39	5/6	246.4	4.9	345.8	72.9	314.1	38.2
R40/42	8/11	32.3	9.9	355.6	69.3	334.6	18.0
R41	6/6	85.5	7.3	11.1	72.0	337.5	23.1
R43	6/6	62.8	8.5	344.8	85.7	324.9	32.0
R44	4/6	62.7	11.7	322.7	79.9	323.0	26.0
R45	6/6	218.7	4.5	8.9	70.7	337.8	21.7
R47	4/6	36.4	15.4	0.2	76.9	5.2	32.0
R49	4/5	83.1	10.1	349.1	79.6	298.7	52.7
R50	6/6	77.6	7.7	334.6	73.0	302.1	45.6
R51	6/7	53.2	9.3	44.6	78.0	305.5	62.5
Mean of site means (N=32)				2.3	76.6	330.8	42.1
(k/α_{95})				(97.7/2.6)		(16.8/6.4)	
Expected directions (Cratonal North America)							
Present Field				23	74		
Bruhnes Field				0	74		
40 Ma				351	76		

For locations and localities of sites, see Table 3 and Figure 2; N/n: N is samples used in analysis and n is total samples collected at site; k: is the *Fisher* [1953] dispersion parameter; α_{95}: 95% confidence cone about mean direction [*Fisher*, 1953]. N/A means not applicable (insufficient n). Dec is declination and Inc is inclination in degrees.

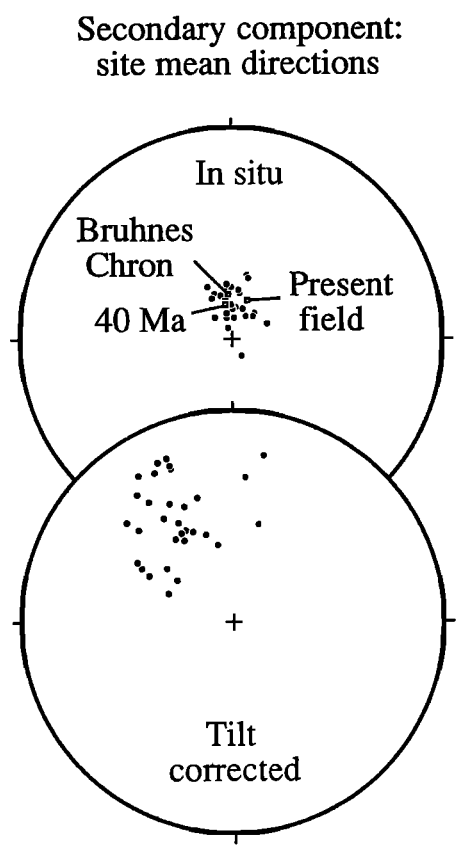


Fig. 10. Stereonet plot of secondary component site mean directions from samples treated with stepwise demagnetization. The data conclusively fail a fold test. The *Fisher* [1953] precision parameter k decreases from 97.7 to 16.8 when the tilt correction is applied (Table 1). The uncorrected direction does not differ significantly from either the average Bruhnes Normal-Polarity Chron or expected middle Tertiary direction.

(Table 1). The mean direction (with no structural correction) does not differ significantly from the expected Brunhes Normal-Polarity Chron direction or the expected middle Tertiary direction (35 Ma [*Diehl et al.*, 1983]), and the data convincingly fail a fold test; The *Fisher* [1953] precision parameter k decreases from 98 to 17 when corrected for tilt (Table 1 and Figure 10).

The high and variable temperatures to which the secondary component persists appear incompatible with a purely viscous overprint acquired during the Brunhes Normal-Polarity Chron. A postaccretion burial CRM could have contributed to the overprint, but the variability of the overprint as well as its absence in the dike samples argue against a widespread CRM. A partial thermoremanent magnetization (PTRM) provides a possible hypothesis. There are circa 35 Ma plutons present in a restricted region of Prince William Sound [*Lanphere*, 1966], but these plutons lie several tens of kilometers away. We hypothesize an origin for the overprint which is dominantly viscous but suppose that phase splitting during thermal demagnetization increases the unblocking temperatures of the secondary remanence by varying amounts.

Characteristic Component

Thermally demagnetized samples . Samples gathered in the recent study were gathered from sites of 4-8 cores and treated with stepwise thermal demagnetization. On orthogonal vector diagrams, the vast majority of samples exhibited a characteristic component with a simple, straight-line decay to the origin defining a direction that was easily determined with a least squares fit to three or more steps. We also observed three other types of behavior during demagnetization: (1) a curved trajectory during decay to the origin (Figure 11*a*), (2) a trajectory that misses the origin (Figure 11*b*), and (3) erratic jumps in direction (Figure 11*c*). We analyzed samples exhibiting the first two types of behavior using great circle demagnetization paths, but only in sites which lack sufficient stable endpoint data to obtain a reliable mean ($n < 3$ samples). All five sites analyzed using great circles came from the Good Cove locality. We discarded samples with erratic behavior.

Directions obtained from individual samples were combined to obtain a site mean direction. Directions from sites with only stable endpoint data ($n = 52$) were combined using Fisher

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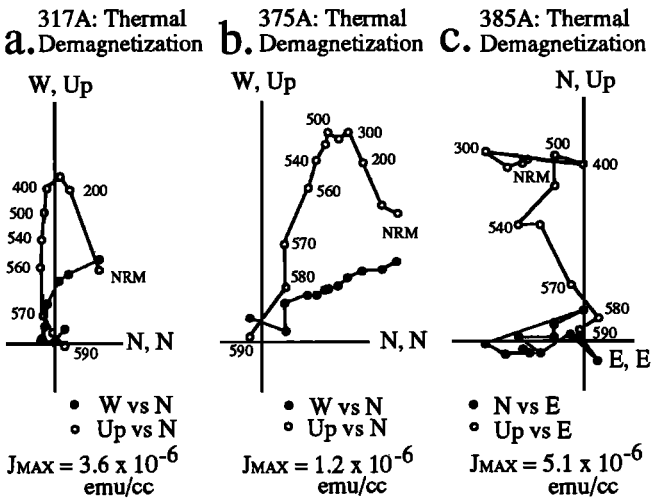


Fig. 11. Orthogonal vector diagrams (in situ coordinates) from three samples of pillow basalt that behaved poorly during stepwise thermal demagnetization. The demagnetization path (*a*) curves into the origin, (*b*) misses the origin, and (c) behaves erratically. Samples in Figures 11*a* and 11*b* were analyzed using great circles. The sample in Figure 11*c* was rejected. Note that all samples appear to have a characteristic component that is steeply up.

statistics [*Fisher*, 1953], while directions from sites that included some demagnetization circles ($n = 4$) were combined using the method of *McFadden and McElhinny* [1988]. We then screened data for reliability using the following criteria.

Within site scatter: We rejected four sites with $k < 20$ and $\alpha_{95} > 20$ (Figure 12). All four sites were from pillow basalt on the south shore of Humpy Cove.

Aberrant directions: One site (R23) has a direction that lies more than two standard deviations from the mean of all site means and was discarded. We had no a priori reason to reject this site and speculate that it records an excursion of the geomagnetic field.

Site grouping: Pillow basalt localities commonly lacked clear indications of elapsed time between successive flows (e.g., interpillow sediments are absent). So as not to bias the mean direction toward flows erupted rapidly in succession, and to estimate the uncertainty more accurately, we combined sequential sites with similar directions (i.e., cannot reject the hypothesis of a common mean with <95% confidence [*McFadden and Lowes*, 1981]) and recomputed their mean. In this fashion, 12 sites were combined in groups of 2-3 sites and recomputed as five sites. The same test was applied to dike sites, but none were combined as a result.

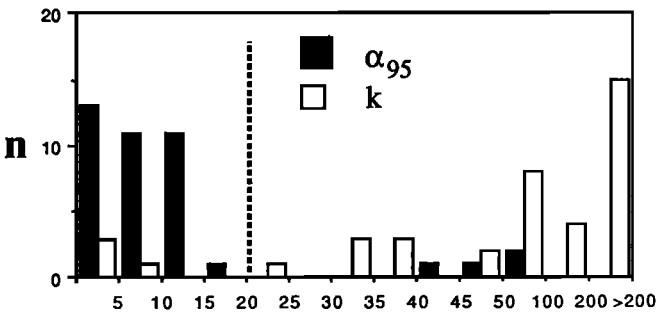


Fig. 12. Histograms of site mean α_{95}s and precision parameter k [*Fisher*, 1953] for all sites with multiple samples (all localities except Killer Bay and Cape Resurrection). Sites with $\alpha_{95} > 20$ and $k < 20$ were rejected (dashed line delineates cutoff). Four sites (R1-R4, Humpy Cove pillow basalt) were discarded by these criteria. Note change in scale for values > 50.

Poor paleohorizontal control: We discarded all sites from dikes at Humpy Cove (six sites) due to uncertainty over their paleohorizontal. Because the pillow basalt and sheeted dikes in Humpy Cove lie in fault contact, we took attitudes from a narrow zone of pillow basalt trapped within the sheeted dikes. These pillows were nearly spherical, only a small number of attitudes were collected, and there is a strong possibility of measurement bias toward surfaces that appeared perpendicular to the enclosing dikes. Because the resulting paleomagnetic direction is very sensitive to the attitude of the paleohorizontal, we considered it prudent to exclude these data (though, overall, they conform to the rest of the data).

AF Demagnetized samples . In the preliminary data set of *Hillhouse and Grommé* [1977], samples from the Killer Bay and Cape Resurrection localities were collected one core per dike and demagnetized with 2-3 steps of af demagnetization. Due to the small number of demagnetization steps, orthogonal vector diagrams help little in analysis. To determine a stable endpoint, we compared the last two steps of the demagnetization and discarded samples if their directions differed by more than 5°. Fifty sites were excluded in this fashion.

There were many dikes sampled at each locality, many of them very close together. In keeping with the combination of successive flows and dikes whose directions do not differ significantly (above), we combined directions from successive dikes that differed by less than 10°. Ten sites were combined into four in this fashion.

A much smaller data set remains after this screening of the earlier data set. Although the criteria for screening do not have a rigorous statistical basing, the resulting means are almost certainly more reliable and the computed confidence limits a more accurate reflection of the true uncertainty.

Structural Correction

The structure of the Resurrection Peninsula ophiolite is relatively straightforward (Figures 2 and 13 and Table 2). Bedded rocks form a west dipping homocline, with dips ranging from 28° to 54° (commonly 30°-40°). The pillows strike N50°E at the northern end of the field area; the strike gradually rotates counterclockwise to N10°W at the southern end of the field area. The dikes are all nearly vertical and strike N54°W in the north and N81°W in the south; also gradually rotating counterclockwise [*Tysdal et al.*, 1977]. Thus the overall structure is an open fold, plunging moderately to the west. Attitudes used for structural corrections are generally well constrained and are taken from the mean of a number of attitudes at each locality. We obtained paleohorizontal at two pillow basalt localities (Good Cove and Hat Cove, Figure 2) from interbedded, thin, silty turbidites. At the other pillow localities, we rely on averaged attitudes from pillows with high aspect ratios. Paleohorizontal for the dikes was obtained from the bedding of either pillow screens within the dikes (Cape Resurrection) or from nearby pillow basalt flows (Killer Bay and Spud Lake).

The structural correction of paleomagnetic data from a plunging fold is not straightforward and requires some careful thought and experimentation. The structural interpretation most commonly used in paleomagnetism is that all folds form

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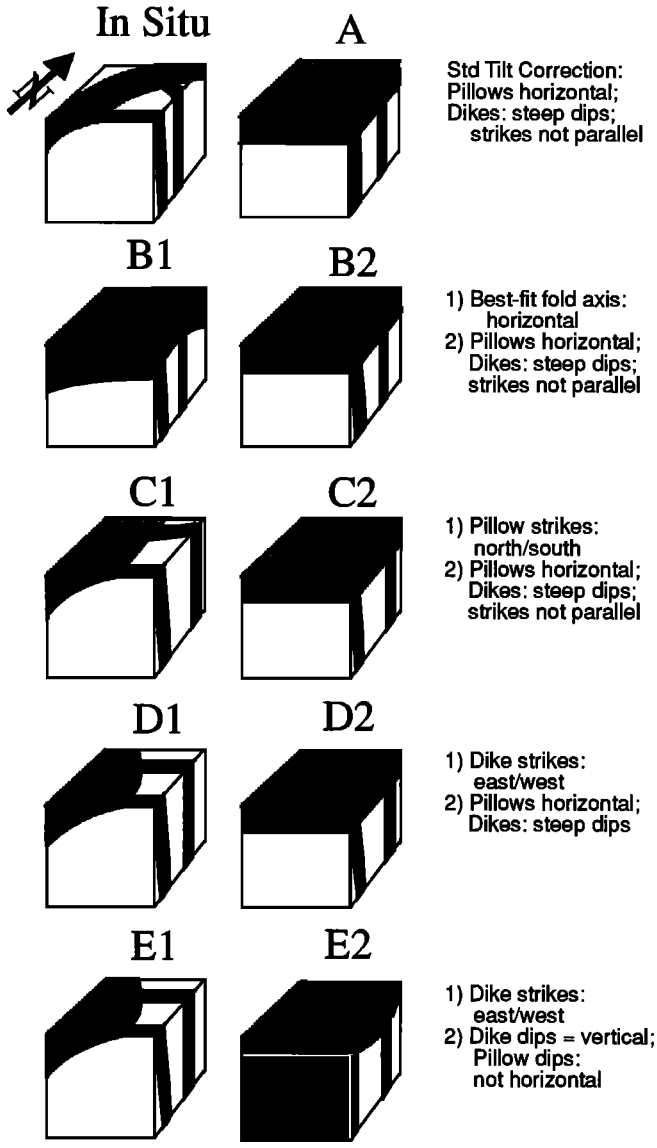


Fig. 13. Schematic block diagrams illustrating the different structural corrections employed in this study. Pillows are represented by the stippled pattern; dikes are black. Figure 14 and Table 3 present corresponding paleomagnetic results.

TABLE 2. Resurrection Peninsula Ophiolite Structural Data

Locality (Lat°N/Lon°E)	Pillow Basalt Strike, Dip	n	$ß_{95}$ $∂_{mn}$ $∂_{mx}$	Sheeted Dikes Strike, Dip	n	$ß_{95}$ $∂_{mn}$ $∂_{mx}$
Spud Lake 59.98/210.74	224°,34°NW	19	3.7 3.8	306°,86°NE	59	3.1 3.2
Cape Res. 59.89/210.73	170°,42°SW	1	-- --	279°,79°NE	46	2.4 2.5
Killer Bay 59.95/210.75	228°,33°NW	11	5.4 5.5	131°,79°SW	4	2.8 2.8
Humpy Cove 59.98/210.71	233°,54°NW	17	3.5 3.6	--		
Porcupine 59.99/210.73	227°,31°NW	13	7.3 7.5	--		
Good Cove 59.92/210.71	184°,30°NW	9s*	4.0 4.0	--		
Hat Cove 59.96/210.70	211°,39°NW	5s*	6.6 6.6	--		
Sandspit 59.94/210.70	193°,32°NW	17	4.5 4.5	--		

Lat°N is north latitude in degrees; Lon°E is the east longitude in degrees; n is number of attitudes; $ß_{95}$ is Bingham 95% confidence ellipse about mean direction [*Onstott*, 1980]; $∂_{mn}$ is minimum axis of $ß_{95}$ ellipse; $∂_{mx}$ is maximum axis of $ß_{95}$ ellipse. ***The s indicates attitudes taken from interbedded clastic sedimentary rocks.**

by tilting about a horizontal axis. If the true fold axis dips more than about 30° and we restore bedding to horizontal by tilting about a horizontal axis (i.e., bedding strike), we introduce a significant error in the declination [*MacDonald*, 1980].

Many recent paleomagnetic studies show declinations that deviate from their expected direction and are interpreted to arise from vertical-axis rotations [*Beck*, 1980; *McCabe*, 1984; *Wells and Coe*, 1985; *Luyendyk et al.*, 1985; *Kissel et al.*, 1986; *Kent*, 1988; *Allerton*, 1989; *Walcott*, 1989]. Studies like these have become increasingly important because geologic studies offer few clues from which we may decipher or quantify vertical-axis rotations. In this study we have both vertical (dikes) and horizontal (sediments and pillows) structural markers that may define a vertical axis fold. We can test this interpretation independently using paleomagnetism if the magnetization predates deformation. Furthermore, we can also draw conclusions about the predeformation structure of the ophiolite.

Below we describe five different approaches to the structural correction (Figure 13). The paleomagnetic results are presented in Table 3 and Figure 14. We compute each of the five structural restorations in two ways: (1) as the mean of all site means (n = 56) and (2) as the mean of eight locality means, each composed of 2-17 sites. We discuss the shortcomings and our preferences of each later in the paper.

The first trial structural correction (A, Figure 13) is the common method of restoring bedding to horizontal by tilting about the strike of bedding. Figure 14 shows that this method results in the four northern localities having a distinctly different declination than the four southern localities. This disparity is due to the difference in strikes, which causes the paleomagnetic directions to "fan out" when untilted.

If the fanning of declinations observed above arises from folding about a plunging axis, then the conundrum becomes what structure most adequately accounts for the observed structural attitudes. A stereonet plot of poles to locality mean bedding and basalt pillow tops shows that we can fit a great circle to those poles fairly well. The pole to this great circle is the best fit fold axis which plunges about 30° to the west (N80°W, 32°W), Figure 15*a*). If we correct the paleomagnetic data by first rotating the best fit fold axis to horizontal and then removing the remaining tilt from each locality (B, Figure 13), we can adequately correct for a simple two-stage deformation that gently tilts a horizontal fold. The paleomagnetic consequence of this restoration shows that, as with the standard tilt correction (A), the northern and southern localities still show distinctly different declinations. Because the best fit axis plunges about 30°, we do not expect results much different from the standard structural correction [*MacDonald*, 1980], as borne out by the results in Table 3.

This best fit fold axis structural correction (B) does not adequately utilize the structural data because it omits the

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TABLE 3. Characteristic Remanent Magnetization, Resurrection Peninsula Ophiolite, Alaska: Site and Locality Mean Directions

					In Situ		Tilt Correction A		B		C		D		E	
Site	Dt	N/n	k	α_{95}	D	I	D	I	D	I	D	I	D	I	D	I
Spud Lake Dikes																
R11	T	6/6	392	3.4	331	-74	120	-71	116	-71	76	-71	85	-71	59	-68
R12	T	6/6	350	3.6	311	-55	250	-88	227	-88	206	-88	215	-88	352	-81
R14	T	6/6	384	3.4	318	-77	131	-70	127	-70	87	-70	96	-70	69	-68
R15	T	4/6	135	7.9	354	-72	103	-67	99	-67	59	-67	68	-67	50	-62
R16	T	2/4	--	--	333	-74	118	-71	115	-71	75	-71	84	-71	59	-67
R17	T	5/5	833	2.7	16	-78	112	-60	108	-60	69	-60	78	-60	62	-57
R18	T	4/6	468	4.3	40	-80	115	-56	111	-56	71	-56	81	-56	67	-53
R19	T	6/6	407	3.3	2	-78	113	-63	109	-63	70	-63	79	-63	61	-59
Mean		8/9	54	7.6	340	-75	116	-69	112	-69	72	-69	81	-69	55	-68
Cape Resurrection Dikes																
6K203	H	1/1	--	--	251	-80	83	-58	87	-58	93	-58	72	-58	78	-59
6K205	H	1/1	--	--	305	-43	358	-58	2	-58	8	-58	347	-58	345	-62
6K210	H	1/1	--	--	296	-85	76	-52	80	-52	86	-52	64	-52	69	-53
6K211	H	1/1	--	--	343	-77	60	-48	64	-48	70	-48	49	-48	52	-51
6K213	H	1/1	--	--	275	-64	58	-72	62	-72	68	-72	47	-72	58	-75
6K216	H	1/1	--	--	273	-48	355	-82	359	-82	5	-82	343	-82	328	-85
6K220	H	1/1	--	--	266	-63	69	-75	73	-75	79	-75	58	-75	72	-77
6K222	H	1/1	--	--	275	-75	72	-62	76	-63	82	-62	60	-62	68	-64
6K223	H	1/1	--	--	309	-69	49	-59	53	-59	59	-59	37	-59	42	-62
6K230	H	1/1	--	--	244	-72	92	-65	96	-65	102	-65	80	-65	89	-65
6K235	H	1/1	--	--	249	-68	92	-69	96	-69	102	-69	80	-69	91	-70
6K237	H	1/1	--	--	231	-77	93	-59	97	-59	103	-59	81	-59	88	-60
6K247	H	1/1	--	--	260	-50	90	-88	93	-88	126	-88	78	-88	153	-86
6K250-1	H	2/2	--	--	249	-63	98	-74	102	-74	108	-74	87	-74	101	-74
6K255	H	1/1	--	--	254	-64	89	-74	93	-74	115	-45	94	-45	98	-45
6K258	H	1/1	--	--	249	-73	87	-65	91	-65	97	-65	76	-65	85	-65
Mean		16/63	28	7.1	270	-69	70	-69	74	-69	80	-69	59	-69	69	-70
Killer Bay Dikes																
6K311	H	1/1	--	--	357	-71	104	-69	96	-69	56	-69	68	-69	81	-70
6K313	H	1/1	--	--	358	-66	90	-69	83	-69	42	-69	55	-69	68	-72
6K315	H	1/1	--	--	292	-59	209	-76	204	-76	161	-76	173	-76	175	-71
6K316-8	H	3/3	3574	2.1	303	-64	178	-80	171	-80	130	-80	142	-80	155	-75
6K320	H	1/1	--	--	322	-66	129	-80	120	-80	81	-80	93	-80	120	-79
6K321	H	1/1	--	--	325	-56	45	-86	33	-86	357	-86	9	-86	132	-59
6K322-3	H	2/2	--	--	322	-67	130	-80	121	-80	82	-80	95	-80	121	-78
6K324-6	H	3/3	754	4.5	348	-71	108	-71	101	-71	60	-71	73	-71	88	-72
6K327	H	1/1	--	--	303	-72	155	-74	148	-74	107	-74	120	-74	133	-70
Mean		9/12	54	7.1	324	-67	127	-79	118	-79	79	-79	91	-79	116	-78
Humpy Cove Pillows																
R8-10	T	20/20	35	5.6	328	-56	135	-70	123	-70	82	-70				
R40,42	T	9/10	94	5.7	352	-47	89	-66	78	-66	36	-66				
R41	T	6/6	89	7.1	338	-51	111	-71	99	-71	58	-71				
R43	T	5/6	367	4.0	320	-59	147	-67	136	-67	94	-67				
R44	T	6/6	250	4.2	341	-72	134	-53	122	-53	81	-53				
R45	T	6/6	287	4.0	319	-68	146	-58	135	-58	93	-58				
Mean		6/10	49	9.7	334	-59	129	-66	118	-66	76	-66				
Procupine Glacier Pillows																
R21	T	4/6	35	15.7	269	-76	163	-66	158	-66	116	-66				
R22	T	6/6	34	11.6	326	-63	96	-84	87	-84	50	-84				
R23*	T	7/7	18	14.5	334	-24	343	-54								
R24-25	T	9/12	53	7.1	320	-70	131	-79	124	-79	84	-79				
R26-28	T	15/18	56	5.1	305	-83	140	-66	135	-66	94	-66				
R29	T	5/5	269	4.7	310	-68	151	-80	146	-80	105	-80				
Mean		5/7	67	9.4	311	-73	144	-76	138	-76	97	-76				
Good Cove Pillows																
R30	T	6/6	28	14.3	307	-82	83	-67	86	-67	79	-67				
R31	T	3/3	1366	5.5	277	-77	102	-70	93	-70	98	-70				
R32	T	4/6	8339	1.9	95	-80	94	-50	95	-50	90	-50				
R33	T	6/6	81	7.5	346	-63	41	-58	43	-58	37	-58				
R34	T	6/6	93	7.5	345	-83	81	-62	82	-62	77	-62				
R35	T	6/6	54	10.9	283	-71	79	-79	81	-79	75	-79				
R36	T	5/6	306	6.4	77	-84	91	-54	92	-54	87	-54				
R37	T	6/7	128	5.9	125	-74	106	-45	107	-45	102	-45				
Mean		8/8	29	10.5	333	-85	85	-62	86	-62	81	-62				

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TABLE 3. (continued)

Site	Dt	N/n	k	α_{95}	In Situ D	In Situ I	Tilt Correction A D	A I	B D	B I	C D	C I	D D	D I	E D	E I
Hat Cove Pillows																
R38	T	4/6	212	6.3	352	-75	95	-58	93	-58	64	-58				
R39	T	6/6	61	8.7	3	-67	80	-55	77	-55	49	-55				
Mean		2/2	--	--	358	-71	87	-57	85	-57	56	-57				
Sandspit Cove Pillows																
R49-50	T	10/11	68	5.9	316	-70	68	-72	68	-72	55	-72				
R51	T	6/7	20	15.3	320	-54	18	-69	18	-69	5	-69				
Mean		2/2	--	--	318	-62	41	-72	41	-72	29	-72				
Means of site means																
Dike sites only (N = 33)					296	-72	93	-74	92	-73	80	-71	72	-71	77	-71
(k/α_{95})					(22/5.4)		(26/4.9)		(29/4.8)		(30/4.6)		(28/4.8)		(25/5.1)	
All sites (N = 56)					310	-74	96	-71	94	-71	78	-69				
(k/α_{95})					(20/4.3)		(24/4.0)		(26/3.8)		(29/3.6)					
Means of locality means																
Dike localities only (N=3)					309	-73	100	-74	99	-73	77	-73	74	-74	73	-74
(k/α_{95})					(44/18.9)		(56/16.5)		(71/14.7)		(222/8.3)		(131/10.8)		(66/15.3)	
All localities (N=8)					322	-72	98	-72	00	-71	70	-70				
(k/α_{95})					(44/8.4)		(37/9.2)		(42/8.6)		(66/6.8)					

Tilt corrections are A, standard structural correction; B, best fit fold axis correction; C, bedding strikes rotated to N-S, then tilt removed; D, bedding strike rotated to N-S, dikes tilted to vertical; E, dike strikes rotated to E-W, then dikes tilted to vertical. Details are given in the text. For location of sites, see Figure 2 and Table 2. Dt is demagnetization technique: T, thermal; H, alternating field. N/n: for site means, N is samples used in analysis, and n is samples collected and measured; for locality means, N is sites used in analysis (after selection and combination, criteria discussed in text) and n is sites collected. Parameter k is the dispersion parameter of *Fisher* [1953]; α_{95} is alpha-95 [*Fisher*, 1953]. D is declination in degrees. I is inclination in degrees.

attitudes of the sheeted dikes (dike localities have both dike and pillow bedding attitudes). Assuming the dikes formed parallel to one another, differences in their attitudes should indicate folding. When sheeted dike and pillow basalt attitudes are solved for a best fit fold axis simultaneously, we obtain a solution by fitting two concentric small circles to the poles of the bedding and dike attitudes (Figure 15b). Because all the sheeted dikes are nearly vertical, we use a vertical axis as an approximate fold axis. We find that a vertical fold axis also fits the pillow basalt attitudes rather well (Figure 15b). For this reason, and the simplicity of rotating data sets about a vertical axis (as opposed to a slightly off-vertical axis), we use

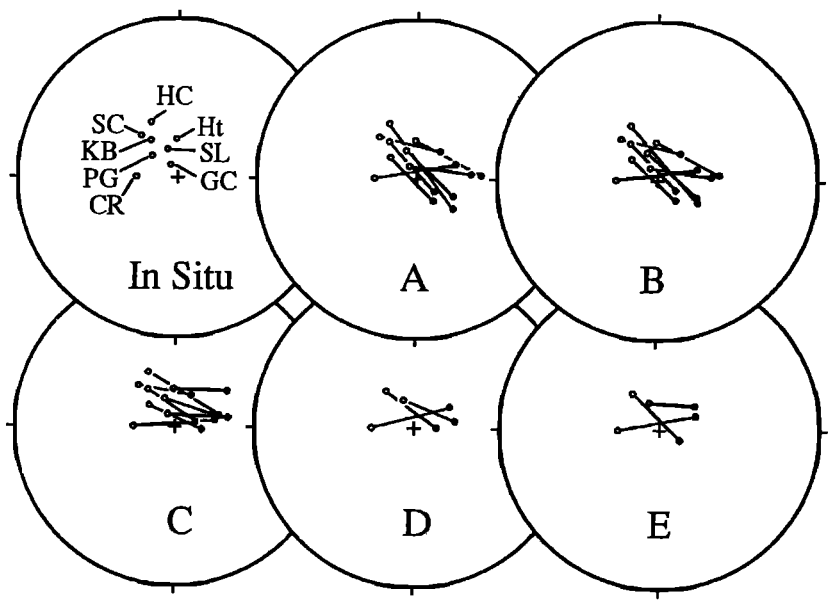


Fig. 14. Equal area stereonet plots of locality mean directions with different structural corrections. All directions are in upper hemisphere. Plot of in situ directions has localities named: CR, Cape Resurrection; GC, Good Cove; HC, Humpy Cove; Ht , Hat Cove; KB, Killer Bay; PG, Porcupine Glacier; SC, Sandspit Cove; and SL, Spud Lake. Locations are shown in Figure 2. Plots A-E show show the movement of locality mean directions when the different structural corrections are applied. These structural corrections are illustrated in Figure 13 and discussed in the text. Note that the directions appear to converge only when corrections C and D are applied.

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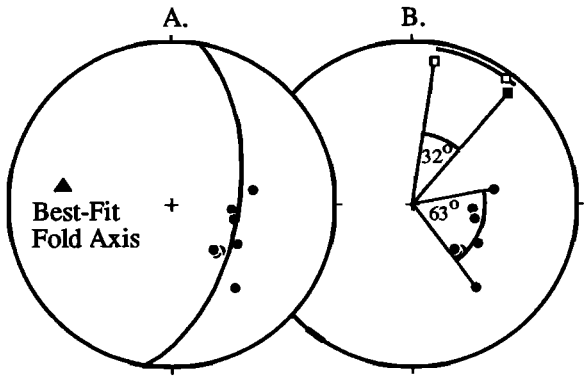


Fig. 15. Two analyses of the structural data. (*a*) Best fit great circle and fold axis to bedding and pillow top locality mean attitudes. The corresponding tilt correction to the paleomagnetic data first rotates the fold axis to horizontal, then rotates bedding to horizontal (about strike). (*b*) Vertical fold axis fit to both paleohorizontal (bedding, pillow tops) and paleovertical (dikes) locality mean attitudes. In this case, concentric small circles are simultaneously fit to both paleohorizontal and paleovertical. The corresponding structural correction first rotates pillow attitude strikes together about a vertical axis, then rotates pillow dips to horizontal. Dike localities all have associated pillow dips, either from pillow screens within the locality or from pillows that lie close by. Paleomagnetic results are discussed in the text and shown in Figure 13.

a vertical axis as one component of each of the next three structural restorations.

We examine three possible compound structural corrections that include one phase of folding about a vertical axis. Because the strike of bedding is rotated about the vertical axis along with the corresponding paleomagnetic direction, the order of rotation does not matter; the paleomagnetic directions arrive at the same spot whether we first rotate about a vertical axis then a horizontal axis or vice versa. Thus it does not matter if we have interpreted the correct order of deformation, just that there are two episodes of folding about nearly vertical and horizontal axes.

The first restoration rotates the strikes of basalt pillow tops and bedding about a vertical axis to N-S (chosen to conform to the overall trend of the peninsula), and then restores the pillows to horizontal (Figures 13 and 14, correction C). Structurally, this correction has the effect of unfolding the plunging fold to a north striking, west dipping homocline and then untilting it. As a result of this restoration, dike attitudes converge somewhat toward E-W strikes, and paleomagnetic declinations from northern and southern localities become nearly identical.

The second tectonic restoration (D, Figure 13) rotates the strikes of the dikes to E-W, then brings the pillows to horizontal. The result of this correction, applied only to the dike localities, does not differ appreciably from restoration (C) above where the pillow strikes were first aligned, but the number of localities is few ($n = 3$).

A third two-stage restoration (E, Figure 13) assumes that the sheeted dikes originally stood vertical and parallel, keeping the line of intersection between dike and pillow attitudes horizontal and allowing the pillows to dip slightly. Paleomagnetic directions converge slightly less than for correction (D), but the difference is not significant because this correction also applies only to the three dike localities.

Interpretation of Structural Restoration

In order to evaluate the various structural corrections we examined two parameters. The first is the well-known k of *Fisher* [1953] which measures the dispersion of a population of directions. A positive fold test, using the above structural corrections, should show a statistically significant decrease in dispersion, i.e., increase in k, [*McElhinny*, 1964]. Using this criterion, only structural restorations which include folding about a vertical axis (C, D, and E) show a decrease in dispersion; the other restorations (A and B) show an increase in dispersion (Table 3). However, none of the results are significant at 95% confidence. In addition, *McFadden and Jones* [1981] have demonstrated that *McElhinney*'s [1964] fold test is inappropriate because, in order to be meaningful, it requires that distributions of data be Fisherian both before and after unfolding.

The second way of evaluating the significance of the various structural corrections is by using a fold test newly developed by *McFadden* [1990]. In it, he examines the correlation of individual paleomagnetic directions with their respective tectonic corrections. The correlation is measured as an angle, small angles suggesting higher correlation, and the sum of the cosines of those angles is computed. When this value (referred to as SCOS) exceeds a critical value for 95% significance, then the paleomagnetic directions can be said to be correlated with the tilting. A significant correlation of directions before structural correction (in situ), but not after, defines a positive fold test and the inverse defines a negative one.

SCOS can be computed according to two definitions. In definition 1, the angle measured is that between the azimuthal direction an individual paleomagnetic direction lies away from the mean direction and the corresponding dip direction of bedding (Figure 16*a*). If the only source of dispersion is tilting, then there should be perfect correlation between the azimuthal direction an individual paleomagnetic vector lies away from the mean and the dip direction (i.e., the angle is zero degrees and its cosine equals one).

For definition 2 (Figure 16*b*), the first element for determining the measured angle is the same: the azimuthal direction an individual paleomagnetic vector lies away from the mean. The second element is the azimuth from the mean direction in the same reference frame as the individual paleomagnetic vector (i.e., in situ or tilt corrected) to the mean direction in the opposing reference frame with the tilt correction applied (the tilt correction that corresponds to the individual paleomagnetic vector). The angle between these two elements is a measure of dispersion, with small angles recording greater scatter.

Definition 1 is appropriate for cases where the mean direction does not move much, relative to the amount of tilt, during structural correction. Definition 2 is appropriate for the opposite case, as in a monocline where all the dips are in the same general direction, and has been applied here.

When applied to our data (Table 4), *McFadden*'s [1990] fold test shows that structural restorations A and B fail a fold test for the mean of locality means as well as the mean of site means. Structural correction C, on the other hand, shows a positive fold test, with >99% confidence, for the mean of site means. The mean of locality means for correction C is inconclusive, however, because the two most discrepant localities, "Hat" and "Sandspit," each comprise only two sites, as compared to a minimum of five and a maximum of 16 sites for the other localities. If these two localities are eliminated, then the fold

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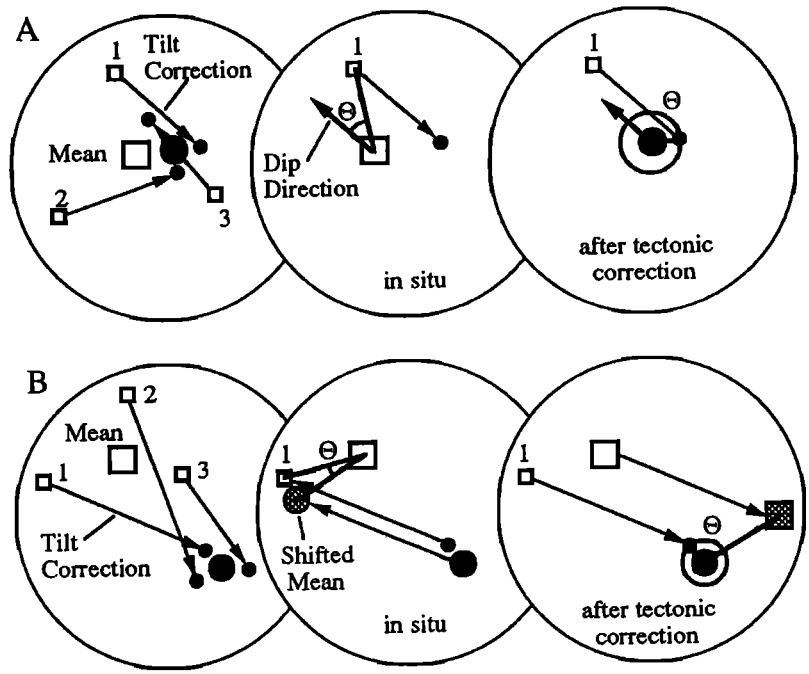


Fig. 16. Illustration of angles used to compute the sum of cosines (SCOS) in *McFadden*'s [1990] fold test using (*a*) definition 1 and (*b*) definition 2. Open squares correspond to in situ directions; closed circles are directions after the tectonic correction is applied. The large symbols represent the mean directions and the small symbols (labeled 1, 2, and 3) represent individual measurements (or site or locality mean directions). Arrows from squares to circles indicate individual tilt corrections. In Figure 16*a* (definition 1), the angle θ is measured between the dip direction and the azimuth from the mean direction to an individual measurement (i.e., from the mean to the point labeled "1") in the same coordinate system (i.e., in situ). In Figure 16*b* (definition 2), the angle θ is measured between the azimuth from the mean direction to an individual measurement and the azimuth from the mean direction to the "shifted mean" (stipled symbols). The shifted mean is the mean direction in the opposing coordinate system with a tectonic correction applied from the applicable individual measurement. In both definitions, a small θ implies a correlation between tilting and the paleomagnetic direction. The cosines for each measured θ are then summed. For small angles (high correlation) the cosine approaches 1 and therefore SCOS approaches the number of individual measurements. SCOS is computed both before and after tectonic correction, and if SCOS exceeds a critical value for 95% confidence, then we may say that the paleomagnetic directions and the tilting are correlated. Significant correlation before tectonic correction, but not after, constitutes a positive fold test. The inverse constitutes a negative fold test.

test for the six remaining localities is positive with >95% confidence. The two remaining structural restorations (D and E) are inconclusive. Conceptually, these corrections are superior to C, because the dikes are constrained to be parallel, but practically, they fail due to the small number of localities (n = 3).

The above results show that paleomagnetic data from the Resurrection Peninsula ophiolite pass a fold test only if we assume folding about both vertical and horizontal axes (C, D, and E). While these corrections do not differ significantly either in their mean inclination or value of k, we use C for the tectonic interpretation below because it encompasses more data.

As mentioned above, we computed means in two different ways: (1) as the mean of all site means and (2) as the mean of eight locality means. Whereas k values are similar for the two methods, α_{95} values differ by a factor of 2-3. The mean of site means has the statistical advantage of more independent measurements of the ancient geomagnetic field. However, it suffers from overly representing the two localities that are the least constrained in quality of data and bedding attitudes (Killer Bay, Cape Resurrection). The mean of locality means emphasizes the best data (detailed thermal treatment) and more equally weights the data geographically and structurally. The α_{95}, while larger, probably represents the true uncertainty more accurately, given the errors inherent in measuring paleohorizontal from the tops of basalt pillows. Thus while some localities are overly represented in the mean (i.e., localities with only two sites), the effect of this bias appears to be negligible; site mean and locality mean inclinations for restoration C differ by less than a degree.

TABLE 4. Fold Test Results Using Test of *McFadden* [1990].

	Structural Correction			Critical Value	
	A	B	C	95%	99%
Mean of All Site Means (n = 56)					
			(+)		
In situ	26.26	26.13	28.37	8.71	12.31
Corrected	29.89	26.05	7.83	8.71	12.31
Mean of Locality Means (n=8)					
		(-)	(-)		
In situ	1.84	1.56	2.02	3.30	4.56
Corrected	6.51	6.39	2.59	3.30	4.56
Mean of Locality Means w/o "Hat" and "Sandspit" (n=6)					
			(+)		
In situ	2.94	2.95	3.09	2.86	3.92
Corrected	4.66	4.46	2.45	2.86	3.92

The table shows SCOS values for each structural correction, using Definition 2 of SCOS. Also shown are 95% and 99% values, which must be exceeded for significant correlation. (+)/(-) denote significant positive/negative fold tests.

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In summary, the mean paleomagnetic direction used in the tectonic interpretation below is the mean of the eight locality means using structural correction C, where the strikes of bedding and pillow tops are rotated to north about a vertical axis before bedding is restored to horizontal.

Error Analysis

The angular dispersion about the mean direction arises from many sources: core orientation during sampling, measurement of paleohorizontal, structural correction, fidelity of magnetic recording, and secular variation of the geomagnetic field. Commonly, we assume these errors and the natural variation are essentially random (nonsystematic); with enough data, they are averaged and are expressed in the uncertainty (α_{95} of *Fisher* [1953]). Systematic errors concern us more because they bias the paleomagnetic pole away from its true value in a way that is not reflected in the confidence limits. Here we analyze possible sources of error in order to determine which are systematic.

Errors arising during orientation and measurement of samples are probably minimal, at most a couple of degrees. We have no reason to believe they are not random and thus greatly reduced in the mean and reflected in the standard error. Orientation mistakes tend to be large and often easily discerned (180° errors are the most common). Other possible error arises from crooked and mismarked cores, but again, the overall effect is probably very small. The TRM of lava flows and dikes is known to be among the most trustworthy in terms of fidelity of recording and preservation of primary signal. The demagnetization experiments indicate a clean isolation of the primary component.

We expect secular variation of the geomagnetic field to compose a significant part of the angular dispersion of magnetic directions at each locality. A larger number of sites at each locality would be needed to ensure a complete representation of this natural variation; two localities have as few as two sites and thus may not provide a true reading of the geocentric axial dipole field. Thus, when we average together all localities, incomplete sampling of secular variation may account for a substantial portion of the site mean dispersion, but any introduced bias should be insignificant and the uncertainty should be reflected in the confidence limits.

The most likely cause of large systematic error is the structural correction. This error arises from two sources: (1) pillow tops might not accurately record paleohorizontal and (2) we possibly selected an incorrect model for unfolding the structure. A single attitude from a pillow top will not be expected to precisely reflect the average of a locality. Therefore we collected a number of attitudes at each locality, usually more than 10 (Table 2). For these attitudes, Bingham statistics [*Onstott*, 1980] reveal ellipses of 95% confidence with axes usually less than 5°, indicating we can be reasonably certain that we have adequately averaged random variation in bedding attitudes. A more likely candidate for systematic error arises from the initial dip of the basalt pillows. To counter this we took attitudes on pillows with high width:thickness aspect ratios (5:1 or better), found high in the flows. We suggest this flattening represents slower cooling pillows that were extruded over the still hot base of the flow, allowing the pillow to sag and "pond" between other pillows. We conclude that these pillows recorded paleohorizontal relatively accurately.

Our two-stage structural correction to restore paleohorizontal remains as the most likely source of systematic error. If every locality had both paleohorizontal and dike (paleovertical) attitudes, we could more confidently correct for both strikes and dips. As it is, we know that results from localities with both dikes and pillows do not differ significantly whether one uses restoration C or D. This similarity indicates that the final result is relatively insensitive to minor variation in the structural correction and that the associated errors contribute no more to the standard error than do other sources.

Overall, we can see that our standard error is the sum of different sources: error measuring paleohorizontal, incomplete averaging of geomagnetic secular variation, and errors stemming from the correction for polyphase deformation. The elliptical, non-Fisherian distribution of locality mean directions possibly indicates a systematic bias arising from inaccuracies in the structural correction that we cannot quantify or undetected variations in the initial dip of pillow basalts. However, an analysis of the observed and expected variance lead us to conclude that errors stemming from variations in initial dip and inaccuracies in the structural correction are relatively small and no greater than errors attributable to incomplete sampling of secular variation and fidelity of paleomagnetic recording and preservation (Appendix A).

Tectonic Interpretation

To reiterate from above, the data used in the tectonic interpretation below come from the mean of the eight locality means using structural correction C (Table 3). While this structural correction yields the shallowest mean inclination (and hence the lowest paleolatitude), we wish to emphasize that it was chosen on structural and paleomagnetic grounds independent of its mean direction. Also note that the mean inclinations (of the locality means) calculated by the different structural corrections differ by only ±1°.

Hagstrum [1990] recently suggested that oceanic rocks are prone to remagnetization during continental accretion, acquiring a CRM induced by fluids migrating out of the subduction zone. He cited as evidence studies which show a remanence of a single polarity which passes a fold test and inferred that the magnetization was acquired during subduction but before accretion, deformation, or appreciable tilting of the descending slab. Thus he believes these studies record the latitude of accretion rather than the latitude of formation. In the present study, we appear to have a similar situation with a single-polarity magnetization that passes a fold test. However, late Paleocene-early Eocene time was characterized by long periods of a reversed polarity geomagnetic field (Figure 17) [*Harland et al.*, 1990]. Therefore rocks which acquired their magnetization over a span of a few hundred thousand years would be expected to show dominantly, if not entirely, reversed inclinations. In addition, in this paper we presented petrographic evidence for a TRM in the diabase dikes and hydrothermal alteration similar to that observed in modern ridge settings, and thus argued that the magnetization in the dikes was acquired during their formation and the magnetization in the pillows was acquired slightly later, during ridge-related metamorphism. The positive fold test supports this contention, and thus we believe we have isolated a primary magnetization.

Cox [1970] observed that a population of paleomagnetic poles usually has a more circular distribution than that of the corresponding directions and hence is better suited to *Fisher*

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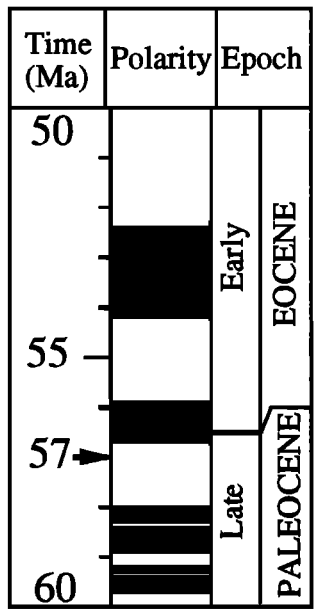


Fig. 17. Geomagnetic polarity time scale for late Paleocene to early Eocene time [*Harland et al.*, 1990]. The long intervals of reversed polarity geomagnetic field are consistent with the reversed polarity magnetization observed in the Resurrection Peninsula ophiolite.

[1953] statistics. Thus we converted each site mean direction (using structural restoration C) to a virtual (instantaneous) geomagnetic pole (VGP). We then combined site mean VGPs into locality mean poles and combined them to obtain a paleomagnetic pole for the Resurrection Peninsula ophiolite at 167°E, 37°N (α_{95} = 11°, k = 28, n = 8). This pole indicates a mean paleolatitude of 54°N±11° at the time the Resurrection Peninsula ophiolite formed.

To interpret the tectonic significance of our data we compare it to paleomagnetic results of the same age from the North America craton. We obtained a 57-Ma pole by interpolating between 50- and 60-Ma poles of *Diehl et al.* [1983]. The resulting pole (186°E, 83°N, α_{95} = 3°) indicates an expected paleolatitude of 67°N for the Resurrection Peninsula. Thus we derive 13°±9° of northward displacement of the Resurrection Peninsula relative to stable North America since 57 Ma. The associated rotation depends on the way the structural correction is carried out; for a pillow basalt strike restored somewhat arbitrarily to N-S (the approximate trend of the peninsula), the rotation is 102°±16° counterclockwise. We computed the 95% confidence limits by the method of *Demarest* [1983]. Taking each locality separately, without rotating the strikes parallel to N-S, the rotations range from 28° to 130° counterclockwise, the lesser rotations corresponding to the more northerly portion of the peninsula.

Northward Displacement of the Chugach-Prince William Terrane

A number of interpretations are possible using this data. Most conservatively, one could insist that with only a few degrees of poleward displacement, the paleomagnetic data are essentially concordant with the reference pole. Based on geologic correlations, the cumulative known or inferred post-mid-Cretaceous displacement on the inboard Tintina and Denali faults is approximately 900 km [*St. Amand*, 1957; *Roddick*, 1967; *Lanphere*, 1978; *Tempelman-Kluit*, 1979; *Gabrielse*, 1985; *Nokleberg et al.*, 1985]. This translates into 5° of poleward displacement [*Plafker*, 1987] and indicates that a displacement value close to that derived from the paleomagnetic data is reasonable.

Circumstantial evidence ties the Resurrection Peninsula ophiolite to the continent soon after formation. Interbedded clastic turbidites within the pillow basalt favor a close proximity to North America. However, turbidites have been known to flow up to 1000 km across a the ocean floor and ascend bathymetric highs [e.g., *Dolan et al.*, 1989], so the tie remains rather loose. Two-mica granite intrusions coeval with the ophiolite (K/Ar ages: 59±2 Ma biotite, 55±2 Ma muscovite)[*Tysdal and Case*, 1979] lie within 5 km, indicating that the ophiolite was perhaps emplaced in the forearc before their intrusion. The contact between the intrusion and the ophiolite is obscured by the waters of Resurrection Bay and the ophiolite lies everywhere in fault contact with the surrounding flysch, so this tie also remains speculative. On Kodiak Island, also in the Chugach-Prince William terrane, similar two-mica granites intrude pillow basalt interbedded with turbidites (Ghost Rocks Formation) [*Moore et al.*, 1983]. Some of those turbidites contain conglomerate, indicating deposition in close proximity to North America. These relationships point to the Ghost Rocks forming close to North America and accreting to it shortly thereafter. We envision a similar history for the Resurrection Peninsula ophiolite and thus interpret the paleomagnetic results to represent the entire Chugach-Prince William terrane.

Faulting between inboard (Peninsular, Wrangellia, and Alexander) and outboard (Chugach-Prince William) terranes most likely explains the discrepancy between displacements observed on the Denali and Tintina faults and that inferred from the Resurrection Peninsula paleomagnetism. The lithology, isotopic compositions, and age of the Orca and Valdez groups in the Chugach Mountains suggest a provenance in the Coast Mountains of British Columbia and southeast Alaska [*Hollister*, 1979; *Plafker*, 1987; *Plafker et al.*, 1989; *Farmer et al.* 1989]. Similarly, J. C. Sample and M. R. Reid (Latest Cretaceous Himalayan scale uplift along the northwest Pacific rim: evidence from sediment volume, sandstone petrography, and Nd isotopic signatures of the Chugach flysch, Alaska, submitted to *Tectonics*, 1992; hereafter referred to as Sample and Reid, submitted manuscript, 1992) report that sedimentologic, petrographic, and isotopic data support derivation of the Kodiak Islands part of the Chugach terrane from the Coast plutonic-metamorphic complex [*Brew and Ford*, 1984] in Late Cretaceous time. While it is possible that coastwise sediment transport may account for much of the apparent displacement [*Plafker*, 1987; *Plafker et al.*, 1989], Sample and Reid (submitted manuscript, 1992) presented paleocurrent data showing radial patterns that appear to discount that theory. As a crude approximation, a restoration correlating the eastern Chugach Mountains (61°N) with the central part of the Coast plutonic-metamorphic complex (54°N) requires 7° of poleward displacement. When added to the 5° poleward displacement estimated for the Denali and Tintina faults, the resulting 12° agrees with early Tertiary paleomagnetic displacements from the Chugach-Prince William terrane (Table 5; Figures 18 and 19).

Paleolatitude and Orientation of the Kula-Farallon Ridge

The mean declination, and hence rotation, found in this study depends upon the structural correction used. The corrected N-S

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TABLE 5. Reliable Late Cretaceous and Early Tertiary Paleomagnetic Results from Southern Alaska.

Study	Terrane*	Age Ma	Poleward Displacement	Lithology and Stability Tests$^{\Delta}$	Reference$^{\Omega}$
Chickaloon Formation.	Wr	56-66	14±11	sf	1
Arkose Ridge Formation.	Wr	56-66	14±11	sf	
Lake Clark	Pen.	56-71	9±11	vfr	2
Alaska Peninsula	Pen	35-45	-8±8	vfr	
Cantwell Formation.	Wr	55-63	0±8	vfr	3, 4
Talkeetna Mountains.	Wr	50-56	-8±10	vfr	5
Talkeetna Mountains.	Wr	39-54	-13±8	vfb	6
Ghost Rocks Formation.	Ch-PW	62-66	25±7		7
Alitak Bay	Ch-PW	62-66	16±10	vf	
Kiliuda Bay	Ch-PW	62-66	33±10	vfr	
Aleutian Islands					
Amlia	Ch-PW	35-45	-7±8	sfr	8
Umnak	Ch-PW	35-45	1±5	sfr	8
Resurrection Peninsula	Ch-PW	57±1	13±9	vf	this study

*Terranes are Wr, Wrangellia; Pen, Peninsular; and Ch-PW, Chugach-Prince William.

$^{\Delta}$Lithology and stability tests are v, volcanic rocks; s, sedimentary rocks; f, positive fold test; r, positive reversal test; and b, positive baked contact test

$^{\Omega}$References are 1, *Stamatakos et al.* [1989]; 2, *Thrupp and Coe* [1986]; 3, *Hillhouse and Grommé* [1982]; 4, *Panuska et al.* [1986, also A revised paleolatitude for the Teklanika volcanics: Evidence for postaccretionary northward translation, submitted to *Journal of Geophysical Research*, 1992]; 5, *Hillhouse et al.* [1985]; 6, *Panuska et al.* [1990]; 7, *Plumley et al.* [1983]; and 8, *Harbert* [1987].

pillow strike and E-W dike strike we used reflect the overall structural trend of the Resurrection Peninsula ophiolite. Therefore the rotation implied by the data is not without meaning and deserves consideration. Several possible mechanisms explain the counterclockwise rotation expressed by our data, reflecting local and/or regional tectonics. Systematic counterclockwise rotations of paleomagnetic data from southwest Alaska have been interpreted to reflect oroclinal bending [e.g., *Coe et al.*, 1989] or terrane translation around curved fault systems [*Hillhouse et al.*, 1985]. The sense of rotation expressed by this study agrees well with data taken further north and may reflect the same processes. However, the position of the Resurrection Peninsula ophiolite within a vast accretionary complex and the possibility of local rotations preclude any confidence in an interpretation of the rotation data.

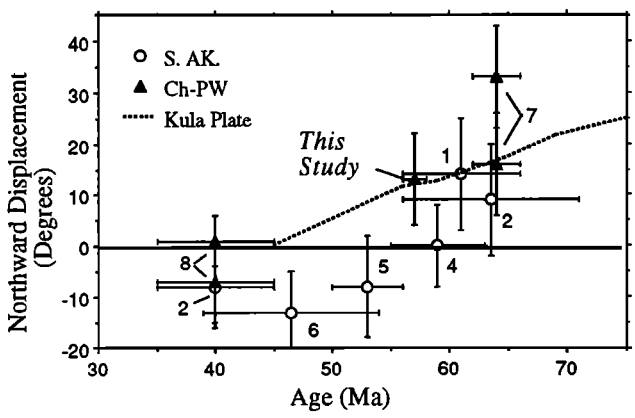


Fig. 18. Displacement versus age for reliable Tertiary paleomagnetic studies from southern Alaska (data and references in Table 5). Although displacement estimates from the Chugach-Prince William terrane are roughly concordant with results from more inboard terranes, they are consistent with a later arrival time, as suggested by sedimentologic, petrographic, and isotopic studies which indicate a Coast plutonic-metamorphic complex provenance for the Chugach-Prince William terrane [*Hollister*, 1979; *Plafker*, 1987; *Plafker et al.*, 1989; *Farmer et al.*, 1989; *Sample and Reid*, submitted manuscript, 1992]. Kula plate trajectory shown (dotted line) is for coast-parallel transport of a point which arrived in southern Alaska (60°N) at 45 Ma [from *Debiche et al.*, 1987]. This agrees well with data from the Chugach-Prince William terrane. Paleomagnetic and geologic results from inboard terranes suggest an earlier arrival, approximately 55 Ma. The discrepancy between inboard and outboard terranes has not yet been accounted for by known faulting. Symbols are S AK, southern Alaska terrane and Ch-PW, Chugach-Prince William terrane.

On the other hand, the declination data are directly related to the strike of the sheeted dikes. The paleostrike of a dike is the difference between the dike's strike azimuth and paleomagnetic declination, after structural correction. The mean paleostrike of the dike complex trends NNE (N26°E±12°, Appendix B). *Moore et al.* [1983] proposed that the Ghost Rocks volcanic rocks formed in response to subduction of the Kula-Farallon ridge in Paleocene time. If the Resurrection Peninsula ophiolite similarly represents a fragment of the ridge, then our data constrain its latitude and orientation.

Circumstantial evidence supports this accreted ridge hypothesis. The dike paleostrike (N26°E) is orthogonal to the spreading azimuth (N64°W) calculated for Kula-Farallon plate motion [*Engebretson et al.*, 1985]. Geochemical constraints favor a MORB-type source for both the Ghost Rocks volcanic rocks [*Moore et al.*, 1983; *Hill et al.*, 1981] (M. R. Reid, Near-trench calc-alkaline and tholeiitic magmatism, Kodiak Island, Alaska: Nd isotope and trace element evidence for origin by early Tertiary ridge-trench encounter, submitted to *Journal of Geophysical Research*, 1992) and the Resurrection Peninsula ophiolite [*Tysdal et al.*,1977]. Furthermore, subduction of a spreading ridge could account for the anomalous, coeval, near-

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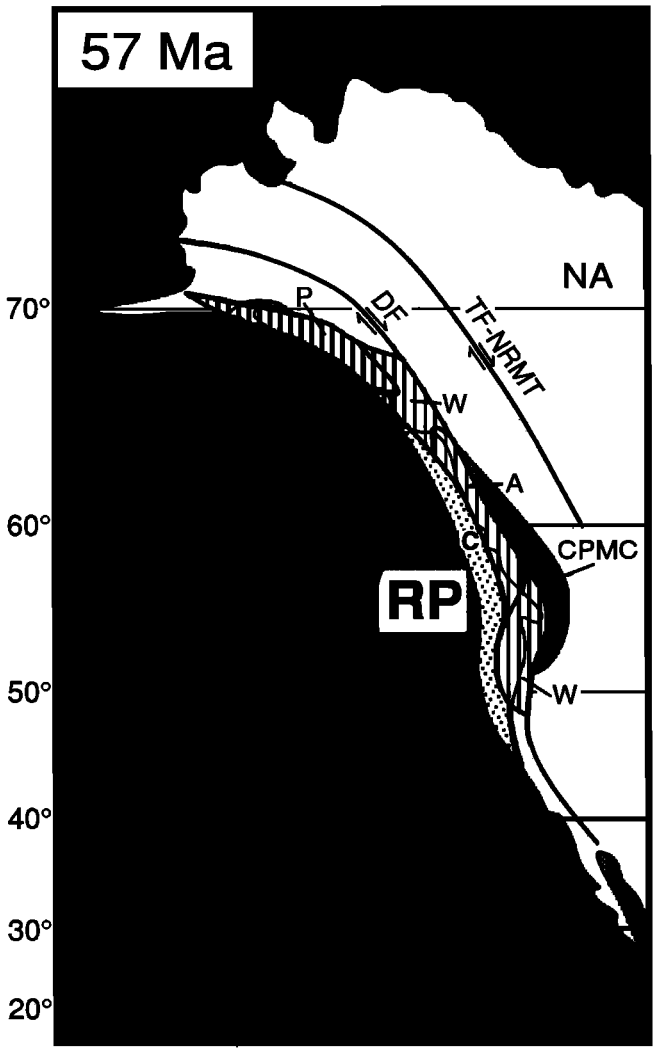


Fig. 19. Tectonic scenario for the northern Cordillera at 57 Ma. The Kula Farallon ridge trends NNE and intersects the continental margin near northern Washington. The Chugach-Prince William terrane lies 13° south of its present position and adjacent to the Peninsular-Wrangellia-Alexander terrane (southern Alaska terrane), shown 5° south of its present position. This implies 8° displacement between the inboard and outboard terranes. The Coast plutonic-metamorphic complex has been shifted south 3°, consistent with 450 km displacement on the Tintina fault-Northern Rocky Mountain Trench [*Roddick*, 1967; *Tempelman-Kluit*, 1979; *Gabrielse*, 1985]. This reconstruction is consistent with (1) isotopic and petrographic studies suggesting a Coast plutonic-metamorphic complex provenance for the Chugach-Prince William terrane [*Hollister*, 1979; *Plafker*, 1987; *Plafker et al.*, 1989; *Farmer et al.*, 1989; *Sample and Reid*, submitted manuscript, 1992], (2) 5° cumulative poleward displacement for the combined Denali and Tintina faults since the Late Cretaceous [*Plafker*, 1987], and (3) paleomagnetic studies suggesting that the inboard terranes arrived slightly earlier than the outboard terranes (Figure 16 and Table 5). The position of the Kula-Pacific-Farallon triple junction is from *Engebretson et al.* [1985], and the Alaska orocline has been straightened to allow the northward migration of terranes in the early Eocene. The dashed ridges show the 95% confidence range for the paleolatitude and trend of the ridge (assuming it formed close to North America). Map Abbreviations are RP, hypothetical position of Resurrection Peninsula ophiolite; A, Alexander terrane; C, Chugach-Prince William terrane; CPMC, Coast plutonic-metamorphic complex; DF, Denali fault; NA, North America; P, Peninsular terrane; TF-NRMT, Tintina fault-Northern Rocky Mountain trench; and W, Wrangellia terrane.

trench, granitic intrusions [*Moore et al.*, 1983; *Marshak and Karig*, 1977]. *Forsythe et al.* [1986] discovered a remarkably similar modern analog where the Chile Rise intersects the Peru-Chile trench.

The position and orientation of the Kula-Farallon ridge through time is a major unknown in Cordilleran tectonics. Plate tectonic models consistently show the Kula plate moving rapidly northward in Late Cretaceous to early Tertiary time [*Engebretson et al.*, 1984, 1985; *Stock and Molnar*, 1988; *Lonsdale*, 1988]. Accordingly, large northward displacements often are explained by invoking rapid spreading across the Kula-Farallon ridge [e.g., *Page and Engebretson*, 1984; *Debiche et al.*, 1987; *Umhoefer*, 1987]. Since the position of the ridge is not known (nearly all oceanic magnetic anomalies having been consumed) and given the possibility of large offsets along fracture zones, the intersection of the Kula-Farallon ridge and North America is unspecified. If we interpret the Resurrection Peninsula ophiolite to be an accreted fragment of the Kula-Farallon ridge and constrain its origin near the continental margin, then during the early Eocene time the ridge lay close to North America near the present Olympic Peninsula (±9° of latitude) and trended NNE (Figure 19).

Recent plate reconstructions for the Pacific basin by *Engebretson et al.* [1984, 1985] outlined two alternatives for the position of the Kula-Farallon-North America (KFNA) triple junction. The northern option places the KFNA triple junction in southern Washington in latest Cretaceous time. Rapid, oblique Kula-North America convergence then accounts for large-displacement strike-slip faults observed in northern Washington and British Columbia [*Plumley*, 1980]. South of the triple junction, rapid, orthogonal Farallon-North America convergence may be associated with the Laramide orogeny. The southern option places the triple junction much farther to the south, near the latitude of southern Mexico. This option puts the Kula plate, with its rapid northward component of motion, adjacent to much of western North America, which helps to explain large poleward displacements inferred for many Cordilleran terranes.

Paleomagnetic paleolatitudes from the Resurrection Peninsula ophiolite and Ghost Rocks Formation constrain the position of the Kula-Farallon-North America triple junction. For 64 Ma (Ghost Rocks Formation) and 57 Ma (Resurrection Peninsula), the paleomagnetism strongly supports the northern option discussed above. From the inception of the Kula plate at 85 Ma up to 64 Ma, however, the these studies have no bearing on the most preferable option. Because the Kula plate had only a small component of northward motion relative to North America during the interval 65-56 Ma, modeled terrane trajectories [*Debiche et al.*, 1987; *Kanter and Debiche*, 1985; *Umhoefer*, 1987] remain relatively unaffected by the change.

Additional constraints are imposed by Kula plate motions, which determine the arrival time of the Chugach-Prince William terrane in southern Alaska. *Debiche et al.* [1987] modeled terrane transport parallel to the continental margin driven by oblique subduction. With the mean displacement of 13°, the Resurrection Peninsula ophiolite would have arrived near its present latitude at 45 Ma. To arrive at the same time, the Ghost Rocks Formation would have had to originate at the northern confidence limit of its paleomagnetic paleolatitude. If this supposition is true, then the Ghost Rocks

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paleomagnetism would also agree more closely with other paleomagnetic studies from southern Alaska (Table 5).

In summary, we interpret the paleomagnetism of the Resurrection Peninsula ophiolite to indicate that the Kula-Farallon ridge trended NNE (N26°E) and intersected North America near the present position of the Olympic Peninsula at about 57 Ma. After 57 Ma, plate tectonic models indicate that it is possible for the Resurrection Peninsula ophiolite to have moved rapidly northward, with the full coastwise poleward displacement of the Kula plate, and arrived in southern Alaska by 45 Ma. This displacement would have been accommodated along the Denali, Tintina, and unidentified fault(s) between inboard terranes and the Chugach-Prince William terrane.

CONCLUSIONS

The paleomagnetism of the Resurrection Peninsula ophiolite indicates that the Chugach-Prince William terrane was 13°±9° south of its present position, with respect to stable North America, at about 57 Ma. The remanence resides in Ti-poor titanomagnetite in both sheeted dikes and pillow basalt. The sheeted dikes appear to have acquired their characteristic remanence as a primary TRM, while the pillows appear to carry a CRM acquired soon after extrusion. The remanence predates deformation of the ophiolite and indicates that it was deformed by folding about vertical and horizontal axes. After structural correction, the paleomagnetism indicates a mean paleolatitude of 54°±7°N, and a mean paleostrike of N26°E±12° for the sheeted dikes.

We interpret these results, in conjunction with geologic relationships, to indicate that the Resurrection Peninsula ophiolite represents an accreted fragment of the Kula-Farallon ridge that lay off the coast of northern Washington/southern British Columbia and trended approximately NNE in earliest Eocene time. Sedimentologic, petrographic, isotopic, and age data indicate a Coast plutonic-metamorphic complex provenance for the Chugach-Prince William terrane. Thus it appears that there is some relative displacement between inboard and outboard terranes. Paleomagnetic studies from terranes inboard of the Chugach-Prince William terrane (i.e., Peninsular, Wrangellia, and Alexander) are consistent with these results.

APPENDIX A

We attempted to quantify the portion of the error attributable to different sources. The major sources of error include (1) incomplete sampling of secular variation of the Earth's magnetic field, (2) fidelity with which the rocks recorded the ambient field during acquisition of their magnetization, (3) errors incurred as a result of the structural correction, and (4) errors due to variation in the initial dip of the pillow basalt. Secular variation will be reflected in the variance between sites within localities and the variance between localities. Paleomagnetic fidelity will be reflected by the variance between samples within sites. Errors stemming from the structural correction or initial pillow dip variation should be recorded in the variance between localities.

The total variance (σ_T^2) is the sum of the variance between localities and the variance within localities:

$$\sigma_T^2 = \sigma_{BL}^2 + \sigma_{WL}^2 \quad \text{(A1)}$$

where σ_{BL}^2 is the variance between localities and σ_{WL}^2 is the variance within localities.

The variance within a locality is the sum of the variance between sites and the variance within sites:

$$\sigma_{WL}^2 = \sigma_{BS}^2 + \sigma_{WS}^2 \quad \text{(A2)}$$

where σ_{BS}^2 is the variance between sites of a locality and σ_{WS}^2 is the variance within sites. Thus

$$\sigma_T^2 = \sigma_{BL}^2 + \sigma_{BS}^2 + \sigma_{WS}^2 \quad \text{(A3)}$$

We first calculated the average within-site variance. To do so, we determined the mean variance observed for all pillow sites and all dike sites, then divided by the average number of samples per site [*McFadden et al.*, 1992]:

$$\sigma_{WS}^2 = \frac{\sigma_S^2}{n} \quad \text{(A4)}$$

where σ_S^2 is the mean site variance (pillow basalt=182.3 and sheeted dikes=50.4) and n is the mean number of samples per site; n=6 for all sites except CR and KB, where n=1.

Dike sites had a consistently smaller variance per site and were thus treated separately from pillow sites. Within the dike sites, two localities had just one sample per site and thus data were used from the remaining dike site, but the mean number of samples per site (n) was taken to be 1. As stated above, we ascribe the within site variance to the fidelity of the rocks to accurately record the paleomagnetic signal.

To determine the variance due to secular variation at each locality, we subtracted the appropriate mean within-site variance from the observed between-site variance. We then averaged the between-site variance in a weighted mean:

$$\sigma_{BS}^2 = \frac{\sum_{i=1}^{n} \sigma_{BSi}^2 (N_i - 1)}{\left[\sum_{i=1}^{n} N_i\right] - 1} \quad \text{(A5)}$$

where N is the number of sites within a locality.

The variance observed between localities is composed of the variance due to secular variation not averaged within localities, errors in the structural correction, and variations in the initial dip of the pillows:

$$\sigma_{BL}^2 = \sigma_{BLSV}^2 + \sigma_{SC}^2 + \sigma_{PD}^2 \quad \text{(A6)}$$

where σ_{BLSV}^2 is the variance between localities stemming from secular variation, σ_{SC}^2 is the variance due to inadequacies in the structural correction, and σ_{PD}^2 is the variance due to variation in the initial dip of basalt pillows.

To arrive at the variance attributable to secular variation only we use the model of *McFadden et al.*, [1992] and determine the expected total variance due to secular variation at 60° latitude in the time interval 80-45 Ma. The total variance due to secular variation should be the sum of the variance between sites and the variance between localities which we can ascribe to secular variation:

$$\sigma_{SV}^2 = \sigma_{BS}^2 + \sigma_{BLSV}^2 \quad \text{(A7)}$$

where σ_{SV}^2 is the total expected variance due to secular variation.

Thus, to determine the variance due to secular variation between localities, we simply subtract the mean between-site

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variance from the expected total variance due to secular variation. Now all that remains is to go back to (A5) above. We know the observed variance between localities and have determined the portion of that variance due to secular variation. The remainder can be attributed to inaccuracies in the structural correction and variations in the initial dip of the basalt pillows. The results are presented below:

$$\sigma^2_{BS} = 18.3^2$$

$$\sigma^2_{BL} = 15.3^2$$

$$\sigma^2_{SV} = 21.6^2$$

$$\sigma^2_{BLSV} = 11.4^2$$

$$\sigma^2_{SC} + \sigma^2_{PD} = 10.2^2$$

APPENDIX B

To determine the mean paleostrike of the sheeted dike complex, we calculated the mean paleostrike of the dikes at each dike locality and then took a weighted average of the three locality means, as follows:

$$\text{mean paleostrike} = \frac{\sum_{i=1}^{3} \frac{Ps_i\,(n_d n_s)_i}{(\delta_d^2\,\delta_s^2)_i}}{\sum_{i=1}^{3} \frac{(n_d n_s)_i}{(\delta_d^2\,\delta_s^2)_i}} \qquad \text{(B1)}$$

where Ps is the locality mean paleostrike (=mean strike-mean declination), n_d is the number of declination determinations within each locality, n_s is the number of strike determinations within each locality, δ_d is the confidence limit on locality mean declination, and δ_s is the confidence limit on locality mean strike.

Now δ_d is determined after the form of *Demarest* [1983]:

$$\delta_d = C \sin^{-1} \frac{\sin \alpha_{95}}{\cos I} \qquad \text{(B2)}$$

where $C = 0.8$, $\alpha 95$ is the $\alpha 95$ [*Fisher*, 1953] of the locality mean direction, and I is the locality mean inclination, as is δ_s:

$$\delta_s = C \sin^{-1} \frac{\sin \beta_{95}}{\sin D} \qquad \text{(B3)}$$

where $C = 0.8$, $\beta 95$ is the major axis of the Bingham-95 ellipse [*Onstott*, 1980], and D is the locality mean dip of the dikes.

The uncertainty of the mean paleostrike (δPs) is taken as the standard deviation of the mean:

$$\delta Ps = \sum_{i=1}^{3} \frac{(Ps_i - \text{Mean } Ps)^2}{n\,(n-1)} \qquad \text{(B4)}$$

where n is the number of localities ($n = 3$).

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